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***“Does Angel Participation Matter?
An Analysis of Early Venture Financing”***

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**SECOND ANNUAL RESEARCH SYMPOSIUM ON
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Session Three – Venture Capital and Entrepreneurship

Does Angel Participation Matter?

An Analysis of Early Venture Financing

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March 24, 2009

Abstract

We examine the role of angel investors in early venture financing using a new sample of 182 Series A preferred stock rounds. Our sample includes deals in which angels invest alone, those where they co-invest with venture capitalists (VCs), and deals in which VCs invest alone. We find that investor composition is strongly related to control rights, and deals with more angel investors have weaker control rights. Our broad results regarding outcomes and investor selection are consistent with a constrained matching explanation regarding how firms pair with investor types. Among smaller deals, matching is unconstrained, and outcomes are not correlated with investor composition. Among larger deals, VC participation is generally a necessary condition for raising capital, and firms for which this constraint binds experience inferior outcomes.

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I Introduction

Private equity investments of business angels represent the lion’s share of investments in de-novo startups.¹ Yet, due to the paucity of data on angel financing (Fenn and Liang 1998; Prowse 1998), the nature of angels’ investments in startups has largely been characterized based on survey evidence (Freear, Sohl, and Wetzel 2002; Wiltbank and Boeker 2007). For instance, the conventional wisdom is that angels tend to invest in early-stage deals, hold common stock, and exert influence through social networks rather than imposing formal control rights (Ibrahim 2007; Wilmerding 2003).

Our paper makes two primary contributions. First, although assortive matching is an established concept in the VC literature (cf., Sorensen 2007), our paper is the first to demonstrate liquidity constraints in the angel investment market and to suggest that these constraints may adversely affect entrepreneurial outcomes. We find that VC participation is generally a necessary requirement to finance larger deals. Thus, we hypothesize that entrepreneurs requiring larger deals face matching constraints, and those who match best with angel investors alone must accept a second-best match. We find that large deals with a high propensity to match with angel investors alone experience inferior outcomes. As predicted by this hypothesis, this result obtains for large deals, rather than for smaller deals where constraints are not likely to exist.

Second, and relatedly, our paper is the first to find evidence consistent with the predictions that control rights are allocated to the party whose marginal contribution to the project is greatest. We hypothesize that angels and VCs primarily differ on their abilities and disposition to influence firm behavior and investment patience. Consistent with this notion, we find that angel investors generally obtain weaker control rights than do VCs. Several theories predict that this should indeed be the case (e.g., Aghion and Tirole 1994), especially when VCs are able to contribute managerial effort that may improve the firm’s prospects (Casamatta 2003; Hellmann 2002).² Our finding is important because despite this strong theoretical foundation, evidence in the existing literature has been elusive. The

¹Extrapolating from the Survey of Small Business Finance, Fenn and Liang (1998) find that for every one firm that raises a venture capital investment, six raise an angel investment. Similarly, they note that approximately one-third of firms that go public were funded by venture capitalists and two-thirds by angels, and conservatively conclude that there are at least double the amount of angel investments as compared to venture capital investments.

²Hellmann and Puri (2000, 2002) document variance in product market strategy and top management team professionalization of VC-backed and non-VC-backed firms. Kaplan and Stromberg (2004) analyze venture capital contracts and find that investor rights vary with the expected and actual investor effort contribution. Dessein (2005) provides an alternative theory of control rights, attributing investor right allocation to signaling. An alternative, and perhaps complementary, explanation is that angels are more patient than VCs (Jovanovic and Szentes 2007). They may relinquish control to mitigate the entrepreneur’s risk of premature liquidation especially when projects require longer time horizons (Lacetera 2008). Both interpretations are consistent with our premise that angels and VCs invest with heterogeneous control rights in equilibrium, a key finding in our paper. We discuss these alternative theories more in Section VII.

key limitation of existing studies, addressed by our study’s examination of heterogeneous investor compositions, has been the two central contexts in which it has been tested: venture capital alone (Kaplan and Stromberg 2004) and research alliances (Higgins 2007; Lerner and Malmendier 2005; Lerner and Merges 1998).³

Although our results are strong, we note a key disclaimer. Our sample of stock purchase agreements and other legal documents are derived from the electronic records of the now defunct law firm Brobeck, Phleger & Harrison (Brobeck). This sample is comprised of high quality deals, as they are effectively certified by a prestigious law firm. This is supported by their higher incidence of successful liquidity events relative to other firms tracked by Venture Economics. Our angels are also more sophisticated than typical angels.⁴ Our results thus might not generalize to very small or very risky transactions. On the other hand, our focus on these larger deals is also a virtue regarding our study’s applicability to the study of venture capital, as more prestigious VCs limit attention to quality transactions. Indeed, using several measures of prestige, we confirm that the VCs in our sample are more prestigious than others in Venture Economics.

Our data include a rich set of controls, including risk, firm age, deal size, and control rights. With these controls in place, we report four central findings.

1. When firms raise smaller amounts of capital, they do so from either angels alone, VCs alone, or from both angels and VCs. In contrast, when larger investments are needed, VC participation is generally necessary, suggesting that matching is constrained.
2. In Series A rounds, angels almost always take preferred shares. Nevertheless, the presence of angels, either investing alone or alongside VCs, is associated with weaker cashflow and control rights. This result obtains for large and small deals alike.
3. Among smaller deals, angel-only deals have the lowest incidence of failure, and a similar incidence of IPOs and acquisitions. However, many of the surviving (non-failed) angel-only firms are less active.
4. When deals are large, those financed by VCs alone are more successful than those in which angels participate.

³There are few studies that explicitly compare VCs and angel investors, perhaps because identifying samples on the margin of VC and angel investment markets has been difficult. For example, Wong (2002) conditions on angel participation, and Kaplan and Stromberg (2003, 2004) condition on VC involvement. Similarly, analyses of private equity returns by Cochrane (2005), Hall and Woodward (2007), Hochberg, Ljungqvist, and Lu (2007), Kaplan and Schoar (2005), Ljungqvist and Richardson (2003), Moskowitz and Vissing-Jorgensen (2002) and Woodward and Hall (2004) are based on investments by VCs or small privately-held businesses. Wiltbank and Boeker (2007) present survey evidence on angel returns.

⁴The mean investment of \$150,375 made by angels in our sample is large relative to the \$10,000 average outside equity investment made by individuals as reported by Reynolds (2005). Shane (2008) reports that investments over \$200,000 fall in the top one percentile of angel investments.

Finding 1 indicates that in Series A investment rounds, deal size segments the market. In particular, entrepreneurs who wish to raise funds from angels alone can do so only when their capital requirements are modest. This situation may arise if angel investors are liquidity constrained which limits the amount of money they are willing to invest in any particular deal. Alternatively, it may be more difficult to find angel investors if angels do not advertise their presence and / or an entrepreneur’s network is limited. This finding has strong implications for the way in which we conduct our analyses. Although we report characteristics and results for the entire sample throughout the paper, we also analyze large and small deals separately. We then focus on a unified theory that can explain results in both subsamples.

The conventional wisdom that angels make smaller investments than do venture capitalists while obtaining weaker control and cashflow rights (Ibrahim 2007; Wiltbank and Boeker 2007; Wong 2002) is consistent with our findings. Moreover, Finding 2 provides unique evidence that control rights vary with investor composition.

Although alternative theories cannot be ruled out, our results for both small and large deals are most consistent with a simple theory of constrained matching. Founders requiring more investor patience, having greater managerial ability, or having a preference for retaining control match best with angel investors. Founders requiring managerial coaching or faster growth match best with VCs. Firms having some of both characteristics match best with mixed investors. This theory fits in well with our findings regarding control rights.

Importantly, this theory can explain the lack of a link between investor composition and outcomes for small deals (Finding 3), and the negative link between angel participation and outcomes for large deals (Finding 4). In particular, we find that all three matches are frequently observed among small deals (Finding 1). We infer that the matching process is not constrained when capital requirements are modest. Hence, no link to poor ex post performance is predicted because all matches are first best. Within the sample of larger deals, however, the rarity of observed angel-only financing suggests that founders face a constrained matching process in which angel-only financing is unlikely. Founders that would have matched best with angels alone must then settle for a second best match, likely to the nearest alternative, which is mixed investor groups. These mismatched firms might then underperform relative to those with a first best match. An empirical test measuring the propensity to prefer angel-only financing in the sample of large deals supports this theory. Although the constrained matching theory seems to be the best fit for our broad findings, it is important to note that we cannot rule out some important alternative theories, which we discuss in Section VII.

In the next several sections we describe our data and results. Given the sparseness of information about angels in the literature, we also provide an appendix that provides detailed descriptions of our sample firms, deals, and to the extent we are able, investor characteristics.

Section II summarizes a theory of constrained matching. Section III provides an overview of our data. Section IV summarizes our results regarding deal characteristics and control rights. Section V presents the relationship between investor composition and outcomes. Section VI expands our results in the context of our constrained matching theory. Section VII discusses robustness, and Section VIII concludes the paper.

II A Theory of Constrained Matching

We hypothesize that entrepreneurs and investors choose one another through a matching process (see Sorensen 2007 for a summary of evidence). We further assume that entrepreneurs seeking smaller deals (but not those seeking larger deals) are able to match efficiently with their first best investor type (angel-only, mixed, or VC-only). In particular, funding large deals exclusively with angels would require assembling a very large number of investors, as angels invest much smaller amounts than do Venture Capitalists (Wong 2002), which limits the likelihood of this occurring. We find (as does Wong) that large angel backed deals are indeed rare. This suggests that entrepreneurs seeking large amounts of capital face a constrained matching process. Firms that would have matched best with angels alone must match with mixed investor groups that include venture capitalists. These suboptimal matches might contribute to generating inferior outcomes.

In Section A (below), we discuss why some firms might match best with angels and others with VCs. In Section B, we explain how this matching hypothesis interacts with expected control right allocations, which in turn motivates our empirical framework.

A Motives for Matching

Various investors have different abilities and objective functions. For example, there is much evidence that VCs are more actively involved in the firm, and also less patient than angel investors. Some of this evidence (including Kaplan and Stromberg 2004, Lerner 1995, and Baker and Gompers 2003) derives from studies of venture capital firms in isolation.⁵ In contrast, other studies derive indirect evidence based on studies of angels alone (Freear, Sohl, and Wetzel 1994; Freear et al. 2002; Prowse 1998). Wetzel (1983) and Freear et al. (1994) provide survey evidence that angels may have more extended time horizons than VCs. Alternatively, angels' preferences may be difficult to assess and consequently, their

⁵Lerner (1995) and Baker and Gompers (2003) find that VCs are generally active on portfolio company boards. In surveys, VCs also report being actively engaged (Gorman and Sahlman 1989, MacMillan, Kulow, and Khoylian 1988, Sapienza 1992, and Sapienza, Manigart, and Vermeir 1996). The evidence of relative VC impatience is more limited: partnership structures constrain investment horizons (Gompers and Lerner 2000; Jones and Rhodes-Kropf 2004; Sahlman 1990) and, together with the scarcity of VC labor resources, may lead to VC impatience (Jovanovic and Szentes 2007).

behavior is unpredictable. In such a case, entrepreneurs may demand stronger rights as a way to insure against consequences of potential incentive misalignment.

Other reasons why some entrepreneurs might match better with different investors include certification and the possibility of accelerating growth. Megginson and Weiss (1991) and Brav and Gompers (1997) suggest that by attracting high quality VC investments, entrepreneurs are able to certify their firms as having higher quality. This provides benefits later, for example, at the time of IPO, or when later round financing will be needed. Based on a comparison of VC-backed and non VC-backed firms, Hellmann and Puri (2000, 2002) find that VC financing is associated with quicker-to-market strategies. This faster financing is associated with increased professionalism and/or CEO replacement.

These studies support the foundations of a matching model in which some entrepreneurs can benefit more from VC involvement and prefer to match with VCs. Other founders might require patience to achieve longer term goals, or possess superior knowledge regarding their industry and prefer to match with more “hands-off” angel investors.

We conclude that the theoretical foundations underlying a simple matching theory of venture financing is well supported by the existing literature along many dimensions. We add the additional assumption that angels have more limited resources than VCs, and thus cannot fully finance larger deals. This leads directly to the prediction that angel-backed deals should be smaller, on average, than deals backed by VCs. Our second prediction is more subtle. If matching is constrained, in the sense that some entrepreneurs may benefit from matching with angels but their capital requirements dictate that they match with VCs, then these entrepreneurs will be unable to obtain a first best match. This leads to the prediction that such entrepreneurs should, on average, experience inferior outcomes as compared to those that are unconstrained.

B Matching and Control Rights

The theoretical literature explains the allocation of control rights in several ways: One stream argues that moral hazard concerns should lead to an allocation of control rights that minimizes monetary or labor underinvestment by the principal and cash-constrained agent, respectively (Aghion and Tirole 1994; Grossman and Hart 1986). The theoretical literature initially viewed this as indicating that control rights should be allocated to the entrepreneur, as an entrepreneur’s effort allocation would be more difficult to specify upfront. More recent work suggests VCs may receive more cashflow rights as compensation for effort (Casamatta 2003). The financial contracting literature arrives at a similar conclusion, but instead hypothesizes that entrepreneurs enjoy private benefits from control of the project, whereas financiers are solely concerned with financial returns (Aghion and Bolton 1992). Alternatively, Dessein (2005) argues that entrepreneurs may give up formal control rights

as a mechanism to signal congruence of goals. Finally, if a project requires greater patience, it may be efficient for the financier to relinquish some rights as a mechanism to credibly commit to the project in the face of potential alternatives (Lacetera 2008).⁶

Consistent with the prediction that control rights should vary across investor groups in equilibrium, Kaplan and Stromberg (2004) provide some evidence in a study of VCs alone. The authors find that when a VC perceives that they can contribute substantially to the firm, they receive greater cashflow rights. When VCs' involvement with portfolio firms is expected to be more acrimonious (e.g., founder-CEO replacement), VCs receive greater control rights.

These arguments, coupled with the motives for matching described earlier, suggest the following prediction: the presence of angels should be associated with weaker control rights. In other words, the equilibrium choices of control and liquidation rights should reveal information about the optimality of a firm's match to angels alone, VCs alone, or mixed investor types. However, a second prediction also follows: In equilibrium, investor composition should not necessarily predict outcomes, unless the matching process is systematically constrained. Further, the degree of control and liquidation rights in deals in which some entrepreneurs face a limited investor pool should reveal the degree of mismatch.

This discussion motivates a two stage model to test the matching theory. The control and liquidation rights (and the fraction of the firm sold) may serve as empirical instruments for the propensity that a given firm matches best with certain investor groups including angel investors. This in turn permits us, after examining the propensity to match with angel investors in a first stage, to test in a second stage whether constrained / second best matching is associated with inferior outcomes. This leads to a final prediction: after controlling for the degree of mismatch, there should be no relation between investor composition and outcomes. Section VI documents the results of our tests of this prediction.

III Data Sources

We now provide a basic description of our data and associated summary statistics. For interested readers, we provide a more detailed description of our data and its summary statistics in an online appendix. The appendix fully captures information presented here and more, so a reader who completes the online appendix can continue reading from Section IV.A. The additional analysis in the online appendix broadly supports the applicability of our sample to the tests we examine in this paper.

⁶It has long been known that the potential for ex-post renegotiation (hold-up) can lead to ex-ante inefficiencies (Hart and Moore 1988). Rotemberg and Saloner (1994, 2000) demonstrate theoretically that the ability to commit to particular projects can increase incentives to provide effort.

Our data are derived from the electronic records of the now defunct law firm Brobeck, Phleger & Harrison (Brobeck), which was founded in 1926, and served clients primarily in California and the western United States (it also had offices in Austin and the East Coast). The firm’s failure was induced by the Internet bust following a high debt growth strategy, and the firm ceased operations in February 2003. This data has been made available by a partnership with the National Digital Information Infrastructure Preservation Program of the Library of Congress and overseen by a blue-ribbon advisory council, a team of legal and technical experts, one of the authors of this study, and a August 9, 2006 decision by Judge Dennis Montali of the United States Bankruptcy Court, Northern District of California.⁷

We identify 182 Series A funding rounds from 1993 to 2002 with sufficient documentation, and an indication of deal closure.⁸ Our data are collected from Stock Purchase Agreements, term sheets, Voting Agreements, Co-Sale documents, First Refusal Agreements, and Investor’s Rights Agreements.⁹ We combine these data with complete histories and outcomes based on public sources including Lexis-Nexis, Hoovers, SEC filings, the Internet Archive (archive.org), and Thomson Financial’s Venture Economics. We further obtain a record of each firm’s internet presence (from 1996 to present), its press releases, and articles about each firm in the popular press. This allows us to identify liquidations, bankruptcies, acquisitions, IPOs, and major company milestones including strategic alliances, product releases, and subsequent VC investments in the firms. For comparison, 111 of our 182 deals also appear in the Venture Economics database (but without the full information to which we have access).

A Investor Classification and Deal Size

We classify investors in the Series A deals into three major categories – founders, VCs, and angels. Founders are identified using a two stage process: (1) some are identified directly in the documents, and (2) they are otherwise identified using a three step rule. (1) the largest common shareholder is identified as a founder; (2) any other common shareholder holding at least 30% as many shares as the largest common shareholder is also a founder; and (3) any common shareholder holding the position of president or CEO is also a founder. For the 165 companies in our sample for which we have founder data, there were a total of 458

⁷This decision authorized the creation of a *Closed Archive* allowing a significant fraction of these records to be saved. The present work is an experimental project designed to test the feasibility of research under the proposed model. See <http://brobeckclosedarchive.org/index.html>.

⁸We define a Series A funding round as the first round in which preferred stock is issued. Some earlier common-only rounds exist, however we do not focus on them because they are relatively infrequent. Also, controlling for them does not change our study’s results, and we generally do not observe many investors from these initial rounds investing again in the Series A round.

⁹The documents in the Brobeck corpus are very similar to the samples provided by the National Venture Capital Association on their website. See http://www.nvca.org/model_documents/model_docs.html.

founders, leading to an average of 2.78 founders per company. We identify venture capitalists primarily by cross-referencing investor names with those in the Venture Economics database, but we also label as VCs investors with names sharing a common word root with the term “venture”. We label as angels all investors who are not otherwise classified as founders or VCs. This category consists of 2,528 different investors across 144 of our 182 sample firms, and is predominantly composed of individuals.¹⁰

We now summarize how we classify the investor composition of our 182 deals. The top panel of Figure 1 shows the distribution of investor composition for deals in our sample. Thirty-two deals (18% of our sample) rely solely on angel investments, 38 (21%) have only venture capital investment, and the other 112 deals (61%) draw upon both angel and VC investors. These mixed deals have a reasonably uniform distribution in participation by the two groups of investors. When they do participate, VCs systematically invest more money than angels. This suggests that deals involving VC investments are larger, which is indeed the case.

The bottom panel of Figure 1 shows the distribution of deal size, i.e., dollars invested in Series A deals, for three investor composition categories: angel-only, mixed, and VC-only deals. Angel-only deals are smaller (median \$1.12 million compared to overall median of \$3.5 million). Mixed deals are somewhat larger (median \$4.49 million) than VC-only deals (median \$3.53 million). While VC-only and mixed deals are large and small, only 3 of the 32 angel-only deals have sizes above the sample median. We conclude that VC participation is virtually a necessity for larger deals. These characteristics lead us to explore five subsamples in our analysis: three subsamples based on investor composition (angel-only, VC-only, and mixed deals), and two subsamples sorted on size (large and small deals).

B Summary Statistics

Table I displays basic firm and investor summary statistics for our sample, and for deals in Venture Economics (when available). Relative to deals in Venture Economics, our sample has a locational bias consistent with Brobeck’s geographical footprint, with a much higher concentration in California (53% vs 36%), Texas (21% vs 6%), and Colorado (8% vs 2%). We use the Venture Economic Industrial Classifications (VEIC) to identify industries, and we hand-classify the 71 deals from our sample that were not in Venture Economics. Overall, 73% of our firms are in Information Technology, compared to 69% in Venture Economics. Our sample is quite similar to Venture Expert in terms of industry. Regarding timing, our sample over-represents deals in the most recent period of our study. Half of our Series A

¹⁰Individuals invest either directly in their name, or through trusts and other investment vehicles including companies set up for investment purposes. Although we cannot identify friends and family, we note that investors with the same name or ethnic origin as founders appear to be a relatively small proportion of the angels in our sample.

rounds occurred after March 2000 (when the Nasdaq index began its precipitous drop), as compared to 35% in Venture Economics. This may reflect Brobeck’s planned high growth strategy in 2000 to 2002, or an increasing reliance on electronic record keeping. We control for these variables in all of our analyses.

With respect to deal size, the mean investment size for Series A deals in our sample is \$6.14 million, which is smaller than the \$7.15 million in Venture Economics. As discussed above, angel-only deals (\$1.63 million) are statistically smaller than mixed deals (\$7.56 million) or VC-only deals (\$5.76 million). The size difference between VC-only and mixed deals is not statistically significant. The average post-money valuation of firms in our sample is \$14.9 million, as compared to \$24.4 million in Venture Economics. This number is also smaller for angel-only deals (\$6.1 million). In terms of age, our sample is overwhelmingly composed of true start-ups (average age is 1.8 years), which is somewhat young relative to the average age of 3.1 years in Venture Economics. Angel-only deals are roughly three months older than VC-only deals (2.1 years vs 1.8 years), and four months older than mixed deals (1.7 years).

Although milestone data is not available in Venture Economics, most firms in our sample did not achieve milestones prior to Series A funding. Only 13% had products developed, and only 10% were in a strategic alliance. These findings are consistent with the young age of our firms. Overall, we believe our sample is quite representative of firms with Series A deals in Venture Economics along some dimensions such as industry, but are somewhat different in timing, size, age and location. Sample selection could affect our results if, for example, there are systematic differences in control rights across geographical locations, as suggested by Gupta (2000), or changes over time. Thus, we control for these characteristics in all of our later tests.

C Investor Characteristics

Panels B and C of Table I summarize investor characteristics pre-Series A, and after accounting for the Series A financing, respectively. In the online appendix, we further discuss the legal forms used to construct investor variables, as well as more information regarding our sample’s investors. The average firm has 2.78 founders, and smaller firms and firms that have only angel investors in Series A have fewer founders on average. Roughly one quarter of our firms have pre-Series A common rounds, however it is important to note that this variable is not related to the composition of investors in the Series A financing, confirming our broader finding that our results regarding control rights and outcomes are not associated with ex-ante financings including seed rounds.

Panel C confirms our important finding that angels invest in smaller deals: only 3 of the 91 large deals are angel-only deals, compared to 29 of the 91 small deals. On average,

12.8 angels invest in angel-only deals, 4.8 VCs invest in VC-only deals, and a total of 14.5 investors participate in mixed-deals. Also consistent with angels not having as much capital as VC firms, more angel investors are needed to fund an angel-only round. Mixed deals are larger, and thus should be expected to have more investors. In mixed deals, 23.4% of the investment comes from angels, and the balance from VCs.

Approximately 70% of our sample angels appear as individuals in the records. Most of the 10% of investors we categorize as “small company” are also likely to be individuals investing through corporate vehicles for tax and estate-planning purposes. Together, these two categories make up over 80% of the dollars invested by angels in the mixed and angel-only Series A deals. “Companies” and “others” (which include universities, non-profits, and governments) play relatively minor roles.¹¹

Since most of the VCs investing in Brobeck deals (“Brobeck VCs”) are included in the Venture Economics database, we conduct an in-depth comparison relative to the other VCs tracked by Venture Economics. The results are formally discussed in our online appendix, but we summarize them here. Brobeck VCs are weighted toward US-based funds (especially California), early stage funds (but not seed stage funds), and older private equity firms. VC firms investing in Brobeck deals have higher prestige, as measured using many proxies from the existing literature. These include each VC’s IPO rate, funds under management, and centrality.¹² Although the prestige difference between Brobeck VCs and Venture Expert VCs is large, we find far less variation in prestige within the Brobeck sample. VCs in large vs small Brobeck deals do not differ along this dimension, and Brobeck VCs investing in VC-only deals have only slightly higher prestige than those in mixed deals. In unreported tests, we control for VC prestige in all of our key tests, and we find that prestige does not explain any of our results. We conclude that Brobeck’s reputation allowed it to attract uniformly higher quality investors, and that prestige does not explain any of our cross sectional findings.

IV Deal Characteristics

We turn now to deal structure. As before, interested readers should refer to our online appendix for more detail. Table II summarizes the rights associated with the Series A preferred stock based on a review of the closing documents. Gompers (1997) and Kaplan and Stromberg (2003) document that preferred stock is typically differentiated from common stock through superior cashflow rights, voting rights, board representation, liquidation rights, redemption rights, and anti-dilution provisions. Our sample’s variation is indeed along these dimensions.

¹¹See the online appendix for more details.

¹²See Ivanov, Krishnan, Masulis, and Singh (2008), Piskorski and Anand (2007), and (Bonacich 1972), for example, for references.

We classify board seats as being assigned to common shareholders using a two step procedure. First, in many cases, the documents identify which board seats are designated by common shareholders or Series A shareholders. Second, in some cases individuals are designated, and we used a fuzzy name matching algorithm to link these seats to specific investors. As shown in Table II, common and preferred shareholders have roughly the same representation (46% versus 54%) in our sample. Not surprisingly, firms with VC-only and mixed Series A deals have more board seats assigned to preferred shareholders, as do larger deals, likely due to the higher VC participation in those deals.

Turning to cashflow rights, preferred shareholders sometimes have stronger claims in the form of cumulative dividend rights. With regular dividends, an annual payment, often a percentage of investment (generally 8%), is paid conditional on a positive shareholder vote. With cumulative dividend rights, this amount accumulates each year. This clause is present in only 9% of our deals, and never appears in angel only deals. Cumulative dividends are investor-friendly, and provide a strong incentive for an accelerated exit. Its prevalence in VC-only deals is consistent with the notion that VCs are less patient investors.

The variable *Liquidation* is a dummy indicating whether preferred shareholders have special liquidation cashflow rights going beyond their initial investment. A value of zero indicates that, after preferred shareholders receive their initial investment, all remaining proceeds upon liquidation go to common shareholders. When the *Liquidation* dummy takes a value of one, preferred shareholders have cashflow rights beyond their initial investment, and in all cases but two, they share these additional cashflows equally with common shareholders (in the two cases, all remaining proceeds go to preferred shareholders up to a specified cap). This variable's mean indicates that 42% of our sample deals provided strong liquidation rights to Series A investors. *Cap on Preferred*'s mean indicates that 47% of these stricter deals had an upper limit on the amount that can be paid to preferred shareholders. Because many preferred liquidation rights are capped, it is important to note that when the company value upon liquidation is high, preferred stockholders will waive their liquidation rights and convert their stock to common. Also note that the liquidation preference in angel-only deals is much less favorable towards investors, which foreshadows a key result of our paper.

Finally, the table shows that preferred shareholders have the right to redeem their shares at will in about one quarter of our deals, typically after a period of time and usually conditional on a Series A majority or super-majority vote. Such a right would be invoked when a firm is not performing well, and is considered to be an investor-friendly term. Angels investing on their own seldom receive such a redemption right in their stock purchase agreements.

A Determinants of Deal Characteristics

Table III reports the results of probit models predicting the likelihood of investor-friendly cumulative dividend and liquidation rights. Table IV reports the results of an OLS model predicting the fraction of board seats allocated to common shareholders, and a probit model predicting the likelihood of redemption rights. All independent variables are standardized (except dummy variables), and we report marginal effects for all probit models in order to give our reported coefficients simple economic interpretations (how much one standard deviation of a given variable impacts the dependent variable).

Table III and Table IV both illustrate that angel investors are associated with more founder friendly deal terms and here we provide evidence in support of Finding 2. In particular, angel-only status implies an 8% lower likelihood of having cumulative dividends for Series A investors, a 33% lower likelihood of having strong liquidation privileges, and a 22% reduced likelihood of having redemption features. In contrast, a larger VC share of the Series A round is associated with a higher likelihood of both liquidation and redemption rights (significantly so for large deals).

We also find evidence in Table IV that angel-only financings cede 16% greater board control to common shareholders (significant at the 10% level for the whole sample, and almost significant for smaller firms). The table also shows, intuitively, that common shareholders receive greater board control (roughly 12% more per standard deviation) when a smaller fraction of the firm is being sold.

Finally, some additional results emerge in Table III and Table IV. Biotechnology firms are associated with 23% more board control for common shareholders, especially for smaller deals (30% more). This might be due to the more knowledge-intensive nature of this business. We also find that investors in firms that previously announced product releases (i.e., mature firms) are roughly 56% more likely to seek cumulative dividend rights when deals are large. Dividend rights strengthen incentives to speed up the commercialization and exit process, and this may only be effective after the uncertainty associated with product development is resolved. Deals in which Brobeck invested had stronger liquidation rights, suggesting that they either encouraged terms that were more investor-friendly, or they were more likely to invest when terms appeared to be more favorable to investors. Redemption rights appear to be less frequent in deals involving Californian companies.

V Outcomes

In Table V, we provide some descriptive statistics regarding the outcomes of the 182 firms in our sample as compared to 9,902 firms in Venture Economics that recorded an initial financing during the comparable 1993-2002 time period. We further condition the Venture

Economics comparison sample on US-based venture capital investment targets founded after 1967 with the first investment not labeled as buyout, acquisition, other or unknown. Outcome variables of the Venture Economics sample are as reported in that database, but weighted to reflect the distribution of deal origination dates in the Brobeck sample. We identify acquisitions and IPOs in our sample through archival sources such as press releases, as does Venture Economics for their universe of firms.¹³

“Non-exited survival” for our firms is defined as the firm still being an ongoing private concern, and an independent company, as of March 2008. This status is based on the web and other public sources. “Active” non-exited survival means that the firm also issued at least one press release (in Lexis/Nexis) or updated its website (as determined from archive.org) between January 2005 and March 2008 (surviving firms are otherwise classified as “inactive”).¹⁴ Failure for firms in our sample indicates that the firm is not surviving and did not experience a positive liquidity event. Outcome variables for the Venture Economics sample are as reported in that database. Since firm failure is often a silent event, only liquidity events are reported reliably in Venture Economics. Finally, we use Venture Economics to identify which of our firms received a subsequent round of financing involving at least one VC investor.

The success of the firms in our sample closely parallels that for the Venture Economics firms. In our sample, 31% of the firms experience a successful liquidity event (IPO or acquisition), 28% are non-exited surviving firms, and the remaining 41% are failures. While the precise figures shown in Table V for our sample and the Venture Economics data differ somewhat, the only statistically significant difference between them is that the incidence of acquisitions for our sample (26%) is approximately 50% higher than that for the Venture Economics firms (17%). Looking across the different subsamples in Table V, there are a number of statistically significant differences in failure (larger and angel-only companies have lower incidences of failure) and in non-exited survival (angel-only are more likely to survive, though some of this is explained by a higher incidence of “inactive” survival). We will shortly examine these differences more carefully using multivariate regressions.

Finally, note that roughly half of the firms in our sample and three-quarters of those in Venture Economics have a subsequent financing round documented in Venture Economics (and thus involving at least one VC), but for angel-only firms, the fraction is much lower. Since it is known that the majority of companies having an IPO do not have VC backing,

¹³Since we are unable to accurately value companies that have been acquired or continue as private companies, we cannot ascertain investors returns. Thus, we focus on determining the success of the companies in terms of survival and profitable exits, rather than measuring the magnitude of investment returns (Gompers, Kovner, Lerner, and Scharfstein 2007).

¹⁴While it is possible that some firms may wish to “fly under the radar” by not making press releases and not keeping an updated website, we expect that this is unlikely given that visibility has a positive effect on the ability to raise capital in private and public markets.

it is possible that many of the firms in our sample with angel-only Series A rounds simply continue to eschew VC financing over time. However, in unreported multivariate probit regressions, we find that while the incidence of future VC-backed financing is negatively related to angel-only Series A financing, the relationship is not statistically significant upon controlling for other factors. We find that the incidence of future VC financing is higher when the fraction sold at the time of the Series A round is higher (potentially indicating a more capital intensive business), the firm is older at the time of the Series A round, there are no product releases before the Series A round (suggesting perhaps that the firm is not able to self-finance quite as quickly), and the firm is not in the IT industry.

We conduct multivariate probit regressions to further analyze the relationship between investor composition and success. Table VI displays the marginal effects for probit models for the full sample as well as the large and small deal subsamples and it is here that we document support for Findings 3 and 4. In columns one, three, and six, the dependent variable is a dummy set equal to one for firms experiencing an IPO or an acquisition, and zero otherwise. We find that VC-only financed firms are more likely to experience these successful liquidity events, but only in the sample of large deals. The effect is also economically large, as the coefficient has a marginal effect of 46%.¹⁵

In columns two, four, and seven, the dependent variable is survival, which is a dummy variable set equal to one for firms experiencing IPOs or acquisitions, and for firms that are still an ongoing private concern as of March 2008 (i.e., our “non-exited survival” firms documented in Table V). We find that firms financed by angels alone are 33% more likely to survive relative to other firms, and 43% more likely to survive among smaller deals (both significant at the 1% level). However, it is important to note that if VCs are more likely to shut down marginally performing firms, then the survival of angel backed firms and VC backed firms may not be strictly comparable (Jovanovic and Szentes 2007). As an alternative in column five, we thus examine the small deal subsample (where there are angel-only deals) in greater detail, and we adjust our definition of survival based on activity levels, and we reclassify firms experiencing “inactive” survival as failures. We find that angel-only deals are not more likely to survive relative to deals with VC-backing in this specification.¹⁶ This

¹⁵We conduct many additional (not reported) robustness checks to ensure that our categorization of angel investors (e.g. angel investment groups, corporations, and Brobeck itself) does not affect our results. These tests uniformly produce results consistent with those displayed. In addition, we examine the impact of using cutoffs other than 100% when categorizing deals as angel-only or VC-only to see whether our results depend on the strict purity investor identification. Since the lowest VC share in the mixed deal group is 26.3%, there are no “near angel-only deals”. In contrast, there are a number of mixed deals that have relatively minor angel participation (“near VC-only deals”); for instance, there are 62 deals that have at least 95% VC share Series A participation, which is dramatically larger than the 38 that are pure VC-only deals. Hence, there is some clustering near the VC-only extreme. In examining robustness to these less stringent definitions of “VC-only”, we find only minor effects to our results.

¹⁶We also conduct ordered probit regressions based on the assumption that IPO/Acquisitions are more successful than surviving firms, which in turn are more successful than failing firms. We find results that

suggests that while angel-only firms are more likely to survive, this relationship is driven by firms surviving in an inactive state. This might arise if these firms indeed require greater patience, or if angel investors are less likely to force these generally unsuccessful firms to liquidate.

VI Evidence of Matching

To explore whether the inferior performance of large mixed deals relative to large VC-only deals (Finding 4) is due to second-best matching by investors and entrepreneurs, we employ a two stage Probit model. In stage one, we estimate the propensity that a deal will be financed exclusively by angels using the sample of small deals where matching appears unconstrained. In stage two, we use the fitted coefficients from the stage one model to compute the propensity that a firm would have matched with angel-only financing in the sample of large deals where matching appears constrained. We then explore whether this proxy for mismatched deals explains our broader findings regarding outcomes.

We report the results of this analysis in Table VII. The first-stage probit model (Column 1) predicts the likelihood that deals are financed exclusively by angels for the small deal subsample. If investors and firms match efficiently, then the deal terms should indeed be associated with the type of investor, but not with outcomes. These characteristics are also required of valid empirical instruments. Our results support the admissibility of the first stage model. Three of four deal terms significantly predict the likelihood of an angel-only match (Column 1), and none of these variables significantly predicts outcomes (Column 2).

In the remaining four columns we report the results of the Stage 2 analysis. In Columns 3 and 4, we first examine this question under the additional assumption of revealed preference, and we assume that an entrepreneur who would have matched with angel-only financing will always choose a mixed deal as the second-best option (deals financed exclusively by VCs are assumed to not be mismatched). Thus, we compute the propensity of angel-only financing as the predicted value from the stage one model, which is then set to zero for VC-only transactions. We call this first propensity measure “Angel-only Propensity x Mixed Dummy”. We then relax this revealed preference assumption in Columns 5 and 6 by allowing positive angel-only propensity scores for VC-only transactions.

In Column 3, we test whether “Angel-Only Propensity x Mixed Dummy” negatively predicts the IPO or acquisition outcome. The results show strong support for the hypothesis that large deals that should have matched with angels alone experience inferior outcomes. This result is highly significant at the 1% level with a t -statistic of -3.03. In Column 4, we show that the VC-only coefficient indeed becomes insignificant (relative to Column 6

are consistent with those of the probit regressions reported in Table VI.

of Table XIII) when we control for the propensity to be angel-only financed. The size of the VC-only coefficient declines by more than half from 0.46 to 0.22 while the propensity variable remains significant at the 5% level.

The “Angel-Only Propensity” negatively predicts the IPO or acquisition outcome dummy when the revealed preference assumption is relaxed (Column 5). However, this variable alone does not subsume the significance of the VC-only dummy. However, the VC-only dummy is subsumed under the slightly stronger assumption that the influence of mismatch will be more severe for mixed deals. In Column 6, we test this by adding the Angel-Only Propensity x Mixed Dummy cross term. This cross-term is significant at the 10% level, and importantly, subsumes both the VC-only dummy and the angel-only propensity level. We note that the loss of significance for the VC-only dummy and the Angel-Only Propensity level in the presence of the cross term is not just statistical; the former coefficient declines by one half and the latter switches sign (although it is close to zero). Finally, in unreported tests, we confirm that weaker control rights do predict outcomes directly, but only in the sample of mixed deals where the link to angel-only propensity is higher. We conclude that the support for the matching hypothesis is strong.

VII Robustness

There are several potential alternative explanations for our results that we consider. Perhaps the most plausible alternative is that large VC-only deals have a higher degree of quality, and that VCs exclude angels from high-quality deals. This hypothesis is consistent with Finding 4 in which VC-only deals outperform mixed deals, and also explains why we only observe differential outcomes in the large deal sample, where incentives might be stronger. Most important for this alternative is the role played by deal quality, and the ability of VCs to perceive this quality early in the investor selection process, allowing them to potentially exclude other investors.

Two of our findings are less consistent with this alternative explanation than with the matching hypothesis. First, this alternative assumes that VC-only deals are better than all mixed deals and hence is not consistent with our finding that once we control for the propensity to be angel-only financed, then investor composition, and in particular the VC-only dummy, no longer predicts outcomes. Second, one might expect that entrepreneurs have greater bargaining power when deals have higher quality, and hence they should use this power to generate stronger entrepreneurial rights. However, VC-only deals have the *strongest* investor rights (Finding 2). Thus, this explanation must rely on the additional assumption that some entrepreneurs are willing to give up control to maximize the probability of good outcomes. This premise is not implausible: Hsu (2004) finds that entrepreneurs, at times,

accept inferior rights if they believe it will increase success probabilities. Although these findings favor the matching theory, we acknowledge that this evidence is indirect.

Relatedly, it is possible that VC-only deals outperform in the large deal sample because they reflect certification from higher quality VCs, and not higher quality matches or higher firm quality. Higher quality VCs may simultaneously demand more investor friendly terms, and due to their greater abilities, are also able to better guide the project post-financing. Hsu (2004) finds evidence that entrepreneurs will pay more for high status VCs. However, additional tests reveal that Finding 4 cannot be attributed to certification. Based on a number of measures including age, size, successful exit rates of investments targets, and centrality (Piskorski and Anand 2007), we do find that VCs investing on their own have slightly higher prestige than those co-investing with angels. However, the difference in prestige between VCs investing in VC-only and mixed deals is rather small relative to the very large difference in prestige between VCs investing in Brobeck deals relative to the non-Brobeck VCs in Venture Economics (see online appendix). Hence, it is perhaps not surprising that (in unreported regressions) we find that the prestige of the VCs investing in a company is not significantly related to control rights, liquidation rights or outcomes. Moreover, the VC-only variable remains significant when we include various controls for prestige and ability.¹⁷

The poor performance of large mixed deals may also be attributable to “deal messiness”. For example, some VCs report that inexperienced angels may be difficult to deal with, and lead to unpredictable hazards in later financing rounds or at exit (Freear et al. 1994). Inconsistent with this hypothesis, we do not find differences in outcomes in the small deals subsample and it is unclear why deal messiness would afflict large but not small deals.¹⁸

The superior performance of VC-only deals might also arise if some deals require more managerial effort than others, and if VCs seek to disallow angels from free-riding when the cost of their managerial effort is high. This hypothesis is consistent with the stronger control rights we observe for VC-only deals, as greater control might be necessary to leverage the effectiveness of this effort. This greater effort, in turn, can explain the superior outcomes we observe, and can also explain why these outcomes are only superior for larger deals where the incentives to provide effort are greatest. It is important to note, however, that the role played by effort is also consistent with matching as some firms can benefit more from VC effort than others, and thus choose VC-only financing. Hence, it is difficult to test

¹⁷We use a variety of different prestige variables constructed from the statistics in the online appendix, as well as an index based on these measures. The VC prestige for each of the 150 firms’ Series A deals in which there is non-zero VC investment is based on a dollar weighted average or an equal weighted average of the prestige scores for the VCs investing in the deal.

¹⁸Moreover, we might expect that not only the presence, but also the number of angels in a deal would increase its “messiness”. However, we do not find a relationship between the number, or proportion, of angels in a deal and outcomes in either the small-deal or large-deal samples.

whether effort can explain our results in a scenario in which matching does not. Although we cannot rule out this alternative, it is not consistent with our finding that once we control for the propensity to be angel-only financed, then investor composition, and in particular the VC-only dummy, no longer predicts outcomes.

VIII Conclusion

This experimental project stemming from the proposed Brobeck archive provides a unique opportunity to better understand the role of angels in financing startups. Two unique features of our sample make it ideally suited to examine how VCs and angels interact when both groups are active investors in the same group of transactions. First, our sample offers a high degree of heterogeneity in investor composition. Second, entrepreneurs in these deals have a meaningful choice between both investor types. We consider a constrained matching hypothesis, and focus on the relationship between investor composition, deal terms, and outcomes. Our paper makes two primary contributions.

First, although assortive matching is an established concept in the VC literature, our paper is the first to suggest that this matching is constrained, and that these constraints can dramatically affect outcomes. We find that entrepreneurs requiring larger amounts of capital rarely match with angel investors alone despite evidence from a propensity model suggesting they should. This is likely due to angel investors being cash constrained, and so VC participation is a necessary condition for financing larger deals (hence large entrepreneurs face a constrained matching process).

We find that deals that are most likely to have binding constraints experience inferior outcomes. As predicted by the constrained matching hypothesis, this result only obtains among larger deals, where constraints exist, but not among smaller deals where no constraints are likely to exist. It is important to note that our results do not imply that VCs destroy value. Quite to the contrary, VCs provide a valuable alternative because these deals – and most larger deals – would not even be feasible without VC participation.

Second, our paper is the first to find evidence consistent with the prediction that control rights should be allocated to the party whose marginal contribution to the project is greatest, as first hypothesized by Aghion and Tirole (1994). In particular, we find that VCs obtain stronger control rights in Series A financings relative to angel investors, consistent with VC participation in managing the firms they invest in. However, although angels accept weaker control rights, contrary to conventional beliefs, we also find that angels still utilize rather sophisticated contractual instruments even when they invest alone.

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Figure 1: **Angel share of deals by number of investors and dollars.**

The figure depicts how deals vary in their investor composition and size. Top panel: frequency of different investor compositions (fraction of angel investors in deal), unweighted and weighted by investment amounts. Bottom panel: distribution of deal sizes for each category of investor compositions. Both graphs are based on the entire sample (182 firms).

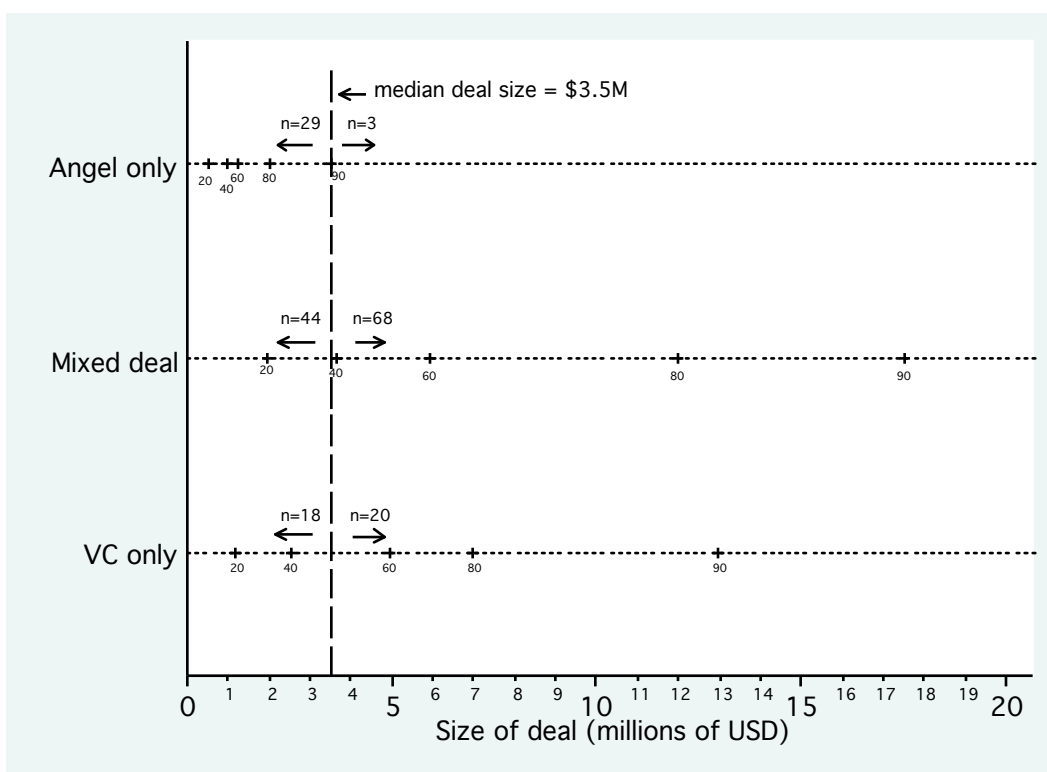
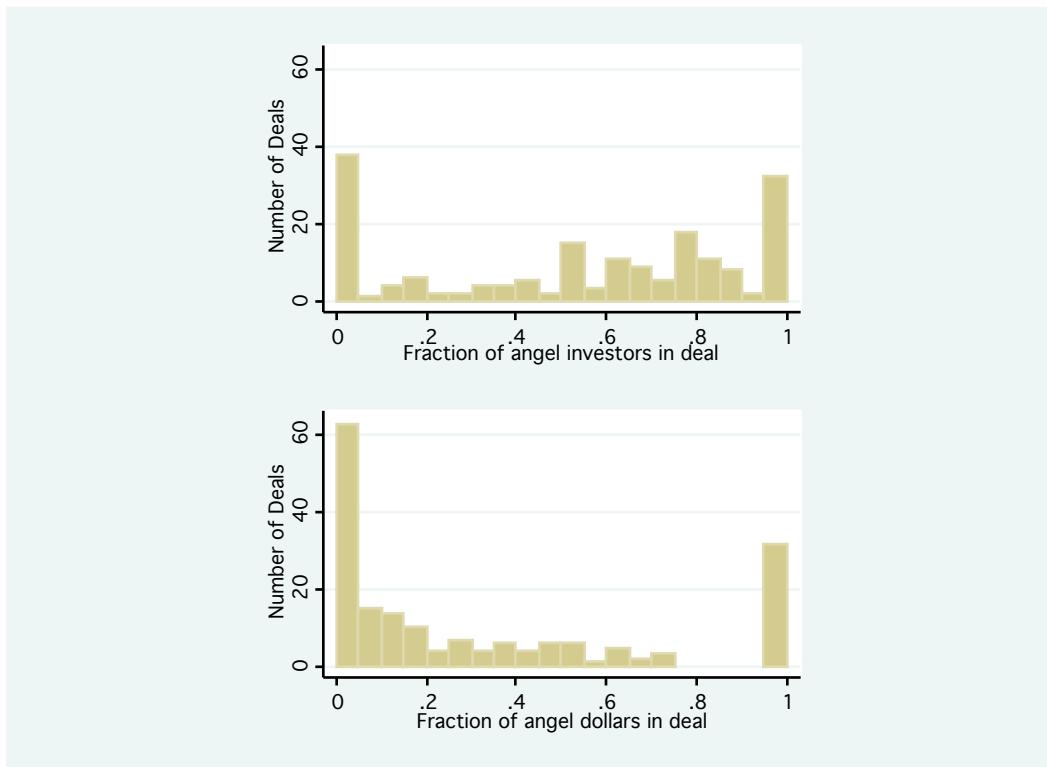


Table I: Summary Statistics

The table displays mean characteristics with standard deviations in parentheses. Demographic information shown in Panel A includes location (state dummies which are based on location of company headquarters) and industry classifications, based upon Thomson Financial Venture Economic's proprietary industrial codes (VEIC). *IT Firm* indicates an information technology firm. *Medical/Bio-Tech Firm* indicates firms classified as "Medical/Health/Life Sciences." For deals that did not appear in Venture Economics, classifications were determined according to Brobeck records and public archival sources. Firm characteristics shown in Panel B, such as time period dummies, are based on the date of first closing. *Size* (and its natural log *Log Size*) is measured based on investment dollars, while *Post-Money Valuation* reflects valuation subsequent to the investment in the firm. *Firm Age* is the number of years between the firm's founding date and the date its Series A preferred stock issue closes (computed as number of days divided by 365.25). *Strategic Alliance* and *Product Release* indicate whether the firm issued a press release describing an alliance or product prior to the observed round. *Company matter* takes the value of 1 if Brobeck did not label the representation "Venture Financing/Investor Side." The Venture Economics sample includes US-based private equity investment targets founded after 1967 with the first investment between 1993 and 2002. Private Equity disbursements labeled as buyout, acquisition, other or unknown were eliminated. Statistics of the Venture Economics sample are weighted to reflect the distribution of deal origination dates in the Brobeck sample, period dummies notwithstanding.

Variable	All Deals	Small Deals	Large Deals	Angel-Only Deals	VC-Only Deals	Mixed Deals	Venture Economics Deals
Panel A: Demographics and Basic Firm Characteristics at time of Series A financing							
California Dummy	0.529 (0.50)	0.587 (0.49)	0.479 (0.50)	0.565 (0.50)	0.423 (0.50)	0.552 (0.50)	0.360 (0.48) ^a
IT Industry	0.731 (0.44)	0.670 (0.47) ^b	0.791 (0.40) ^b	0.750 (0.44)	0.658 (0.48)	0.750 (0.43)	0.692 (0.46) ^a
Medical/Bio-Tech Industry	0.115 (0.32)	0.121 (0.32)	0.110 (0.31)	0.156 (0.36)	0.158 (0.37)	0.089 (0.28)	0.127 (0.33)
1998 to 3/2000 Dummy	0.341 (0.47)	0.429 (0.49) ^a	0.253 (0.43) ^a	0.406 (0.49)	0.289 (0.46)	0.339 (0.47)	0.384 (0.49)
Post 3/2000 Dummy	0.495 (0.50)	0.341 (0.47) ^a	0.648 (0.48) ^a	0.438 (0.50)	0.500 (0.50)	0.509 (0.50)	0.345 (0.48) ^a
Size (Series A Millions)	6.139 (7.92)	1.554 (0.90) ^a	10.72 (9.11) ^a	1.633 (1.53) ^a	5.756 (8.93)	7.556 (8.19)	7.15 (12.15) ^{b,*}
Firm Age (years)	1.795 (4.23)	1.924 (5.03)	1.674 (3.34)	2.069 (4.48)	1.831 (3.60)	1.702 (4.39)	3.14 (4.7) ^{a,†}
Strategic Alliance	0.104 (0.30)	0.066 (0.25) ^b	0.143 (0.35) ^b	0.031 (0.17) ^a	0.105 (0.31)	0.125 (0.33)	N/A
Product Release	0.126 (0.33)	0.110 (0.31)	0.143 (0.35)	0.125 (0.33)	0.158 (0.37)	0.116 (0.32)	N/A
Panel B: Pre Series A Investor Characteristics							
Pre Series A Angel Ownership	0.076 (0.15)	0.076 (0.14)	0.076 (0.16)	0.066 (0.10)	0.091 (0.20)	0.074 (0.14)	N/A
Number of Founders	2.775 (2.46)	2.374 (1.54) ^a	3.176 (3.08) ^a	1.969 (1.06) ^a	2.895 (3.64)	2.964 (2.22)	N/A
Pre Series A Common Round	0.253 (0.43)	0.286 (0.45)	0.220 (0.41)	0.219 (0.42)	0.263 (0.44)	0.259 (0.44)	N/A
Panel C: Series A Investor Characteristics							
Angel-Only	0.176 (0.38)	0.319 (0.46) ^a	0.033 (0.18) ^a	1.000 (0.00)	0.000 (0.00)	0.000 (0.00)	N/A
VC-Only	0.209 (0.40)	0.198 (0.40)	0.220 (0.41)	0.000 (0.00)	1.000 (0.00)	0.000 (0.00)	N/A
Fraction Sold	0.462 (0.22)	0.363 (0.15) ^a	0.561 (0.23) ^a	0.303 (0.16) ^a	0.436 (0.24) ^b	0.516 (0.21)	N/A
% Angel Series A	0.319 (0.36)	0.464 (0.40) ^a	0.174 (0.24) ^a	0.996 (0.01) ^a	0.000 (0.00) ^a	0.234 (0.21)	N/A
% Founder Series A	0.001 (0.00)	0.002 (0.01)	0.001 (0.00)	0.004 (0.01)	0.000 (0.00) ^a	0.001 (0.00)	N/A
% VC Series A	0.679 (0.36)	0.533 (0.41) ^a	0.825 (0.24) ^a	0.000 (0.00) ^a	1.000 (0.00) ^a	0.764 (0.21)	N/A
# Investors	12.12 (13.4)	10.80 (14.4)	13.45 (12.3)	12.81 (20.7)	4.763 (5.62) ^a	14.42 (11.9)	N/A
Brobeck Investor Dummy	0.236 (0.42)	0.165 (0.37) ^a	0.308 (0.46) ^a	0.063 (0.24) ^a	0.079 (0.27) ^a	0.339 (0.47)	N/A
# Firms in Group	182	91	91	32	38	112	9901

* a, b, and c denote significant differences in means at the 1%, 5%, and 10% levels, respectively. The footnotes depicted in columns 2 and 3 are based on tests of differences in means across small versus large firms (column 2 vs 3). The footnotes depicted in column 4 (angel-only) are based on tests of angel-only deals versus mixed deals (column 4 vs column 6). The footnotes depicted in column 5 (VC-only) are based on tests of VC-only deals versus mixed deals (column 5 vs column 6).
 * 9,345 observations; † 3,791 observations; ‡ 9,498 Observations

Table II: Summary Statistics for Cashflow and Control Rights

The table displays mean characteristics with standard deviations in parentheses. *Warrants Dummy* is one if warrants were issued in the Series A Financing. *Options Dummy* is one if an option plan was set up concurrent with the Series A Financing. *# Closings* is the number of separate executed closings under the Series A terms (*Multiple Closings* indicates more than one). *Average Days to Second Closing* is the days between the first and second closings. All preceding variables are taken from the electronic record of the closing documents in the Brobeck corpus. *Billed Hours* reflects the total billed hours associated with Brobeck's representation of either the company or its investors in the deal. Control rights in Panel B and terms are extracted from closing documents from the Brobeck corpus. *% Common BOD Seats* is the share of board seats held by common shareholders (from the "voting rights agreement", or when specific parties are named, by cross-referencing with investors). *Cumulative Dividend Flag* is a dummy equal to one if preferred shareholders have a right to dividends that accumulate over the time of their investment. *Liquidation Flag* is a dummy equal to one when preferred shareholders have liquidation rights exceeding the value of their initial investment. *Cap on Preferred* is a dummy equal to one if these additional preferred stock liquidation rights are capped at a multiple of the initial investment. *Redeemable Flag* is a dummy equal to one when preferred shareholders can demand that the firm repurchase their shares. *Delayed Redemption* is a dummy equal to one if preferred stockholders' redemption rights are time-delayed. *% Series A Req. to Vote* is the percentage of Series A shareholders required to invoke a redemption.

Variable	All Deals	Small Deals	Large Deals	Angel-Only Deals	VC-Only Deals	Mixed Deals
% Common BOD Seats	0.464 (0.33)	0.579 (0.33) ^a	0.359 (0.29) ^a	0.675 (0.31) ^a	0.522 (0.39)	0.398 (0.29)
Cumulative Dividend Flag	0.088 (0.28)	0.044 (0.20) ^a	0.132 (0.34) ^a	0.000 (0.00) ^a	0.158 (0.37)	0.089 (0.28)
Liquidation Flag	0.418 (0.49)	0.374 (0.48)	0.462 (0.50)	0.125 (0.33) ^a	0.579 (0.50)	0.446 (0.49)
Cap on Preferred	0.474 (0.50)	0.441 (0.50)	0.500 (0.50)	0.500 (0.57)	0.364 (0.49)	0.520 (0.50)
Redeemable Flag	0.236 (0.42)	0.209 (0.40)	0.264 (0.44)	0.031 (0.17) ^a	0.368 (0.48)	0.250 (0.43)
Delayed Redemption	0.225 (0.41)	0.209 (0.40)	0.242 (0.43)	0.031 (0.17) ^a	0.368 (0.48)	0.232 (0.42)
% Series A Req. to Vote	57.55 (8.40)	57.12 (8.54)	57.87 (8.52)	50.00 (0.00)	54.16 (7.53) ^a	59.96 (8.35)
# Firms in Group	182	91	91	32	38	112

* a, b, and c denote significant differences in means at the 1%, 5%, and 10% levels, respectively. The footnotes depicted in columns 2 and 3 are based on tests of differences in means across small versus large firms (column 2 vs 3). The footnotes depicted in column 4 (angel-only) are based on tests of angel-only deals versus mixed deals (column 4 vs column 6). The footnotes depicted in column 5 (VC-only) are based on tests of VC-only deals versus mixed deals (column 5 vs column 6).

Table III: Cashflow Rights

The table displays Probit model marginal effects with robust t -statistics in parenthesis. We report results for the whole sample, as well as for small and large deal subsamples as noted in the table header (small and large deals have below and above median deal size, respectively). All independent variables (except dummy variables) are standardized to clarify exposition. The dependent variable in columns 1 and 5 is a dummy variable indicating whether the Series A stockholders have cumulative dividend rights (the cumulative dividend regression cannot be run for the small firm subsample because only 9 firms in this subsample had cumulative dividend rights). The dependent variable in columns 2, 4 and 6 is a dummy variable indicating whether preferred shareholders have liquidation rights exceeding the value of their initial investment. The independent variables include various investor and firm characteristics, as well as industry and time dummies, and are described in Tables I and II.

Independent Variables	<i>All Deals</i>			<i>Small Deals</i>			<i>Large Deals</i>		
	Probit Model	Dep. Var.=	Probit Model	Probit Model	Dep. Var.=	Probit Model	Probit Model	Dep. Var.=	Probit Model
	Cumulative	Liquidation	Liquidation	Cumulative	Liquidation	Cumulative	Cumulative	Liquidation	Liquidation
Angel-Only	-0.08 (-1.67) ^c	-0.33 (-2.49) ^b	N/A		-0.35 (-2.62) ^a				
VC-Only	0.03 (0.53)	0.33 (2.59) ^a			0.24 (1.25)	0.08 (0.87)		0.68 (4.23) ^a	
Pre Series A Angel Ownership	0.01 (0.37)	-0.11 (-2.16) ^b			-0.03 (-0.58)	-0.04 (-0.83)		-0.34 (-3.22) ^a	
Fraction Sold	0.00 (-0.13)	0.01 (0.24)			0.13 (1.41)	-0.02 (-0.45)		-0.03 (-0.35)	
Log Deal Size	0.03 (1.22)	-0.12 (-2.34) ^b			-0.23 (-2.79) ^a	0.02 (0.21)		-0.36 (-2.58) ^a	
Log # Investors	-0.02 (-0.83)	0.07 (1.15)			-0.05 (-0.61)	0.00 (-0.09)		0.33 (2.61) ^a	
Log One Plus Firm Age	0.07 (2.23) ^b	0.06 (1.43)			0.07 (1.26)	0.13 (1.86) ^c		0.23 (2.40) ^b	
Log Number of Founders	0.03 (1.17)	0.08 (1.70) ^c			0.29 (4.00) ^a	0.05 (1.03)		-0.03 (-0.44)	
Pre Series A Common Round	0.04 (0.76)	0.10 (0.94)			0.00 (0.02)	0.02 (0.23)		0.21 (1.31)	
Strategic Alliance	0.02 (0.23)	0.21 (1.44)			0.31 (1.38)	0.25 (1.59)		0.65 (3.15) ^a	
Product Release	0.16 (1.44)	0.24 (1.51)			0.50 (2.51) ^b	-0.09 (-0.48)		-0.31 (-1.21)	
IT Industry	-0.02 (-0.25)	-0.33 (-2.85) ^a			-0.37 (-2.39) ^b	0.01 (0.05)		-0.53 (-2.83) ^a	
Medical/Bio-Tech Industry	-0.06 (-0.81)	-0.08 (-0.53)			-0.21 (-1.37)	-0.04 (-0.22)		0.07 (0.24)	
1998 to 3/2000 Dummy	0.08 (1.85) ^c	0.25 (1.80) ^c			0.28 (1.69) ^c	0.10 (1.23)		0.58 (2.15) ^b	
Post 3/2000 Dummy	0.07 (1.56)	0.35 (2.63) ^a			0.21 (1.12)	0.21 (2.31) ^b		0.69 (3.06) ^a	
California Dummy	-0.02 (-0.37)	0.09 (0.97)			0.03 (0.21)	-0.01 (-0.15)		0.21 (1.31)	
Brobeck Investor Dummy	-0.07 (-1.31)	0.25 (2.42) ^b			0.37 (1.77) ^c	-0.05 (-0.55)		0.29 (1.76) ^c	
Observations	182	182			91	91		91	

* a, b, and c denote significant differences from zero at the 1%, 5%, and 10% levels, respectively.

Table IV: Control Rights

The table displays Probit model marginal effects and OLS model coefficients with robust t -statistics in parenthesis. We report results for the whole sample, as well as for small and large deal subsamples as noted in the table header (small and large deals have below and above median deal size, respectively). All independent variables (except dummy variables) are standardized to clarify exposition. The dependent variable in columns 1, 3, and 5 is the fraction of board seats that are allocated to common shareholders (board of directors data is only available for 147 firms). The dependent variable in columns 2, 4, and 6 is a dummy variable identifying whether the Series A shares are redeemable (i.e., whether they can be returned to the firm in exchange for cash). The independent variables include various investor and firm characteristics, as well as industry and time dummies, and are described in Tables I and II.

Independent Variables	<i>All Deals</i>			<i>Small Deals</i>			<i>Large Deals</i>		
	OLS Model	Probit Model		OLS Model	Probit Model		OLS Model	Probit Model	
	Dep. Var. = Common Seats	Dep. Var. = Redeemable		Dep. Var. = Common Seats	Dep. Var. = Redeemable		Dep. Var. = Common Seats	Dep. Var. = Redeemable	
Angel-Only	0.16 (1.70) ^c	-0.22 (-3.25) ^a		0.17 (1.42)	-0.04 (-3.26) ^a		-0.07 (-0.80)	0.38 (2.24) ^b	
VC-Only	0.04 (0.61)	0.15 (1.54)		0.15 (0.98)	0.00 (0.10)		0.09 (2.98) ^a	-0.18 (-3.56) ^a	
Pre Series A Angel Ownership	0.04 (1.44)	-0.05 (-1.35)		0.01 (0.15)	0.00 (2.97) ^a		-0.06 (-1.60)	0.01 (0.13)	
Fraction Sold	-0.12 (-3.69) ^a	0.01 (0.32)		-0.13 (-1.49)	0.00 (1.54)		0.05 (0.80)	0.00 (0.04)	
Log Deal Size	-0.02 (-0.48)	0.00 (-0.09)		0.03 (0.31)	0.01 (2.65) ^a		-0.04 (-1.06)	0.14 (2.14) ^b	
Log # Investors	0.00 (0.06)	0.01 (0.17)		0.05 (0.62)	-0.01 (-3.46) ^a		0.04 (0.82)	0.12 (1.98) ^b	
Log One Plus Firm Age	-0.02 (-0.60)	0.08 (2.47) ^b		-0.04 (-0.93)	0.00 (3.06) ^a		0.09 (2.89) ^a	0.03 (0.76)	
Log Number of Founders	0.06 (2.58) ^a	0.02 (0.64)		0.05 (1.13)	0.00 (-0.33)		-0.01 (-0.21)	-0.04 (-0.37)	
Pre Series A Common Round	0.04 (0.88)	0.02 (0.27)		0.07 (0.85)	0.02 (2.24) ^b		0.26 (2.45) ^b	0.71 (3.45) ^a	
Strategic Alliance	-0.02 (-0.21)	0.19 (1.53)		-0.23 (-1.08)	0.00 (-0.63)		-0.07 (-0.49)	-0.19 (-1.31)	
Product Release	0.15 (1.63)	0.07 (0.60)		0.11 (0.67)	0.03 (1.88) ^c		-0.08 (-0.85)	-0.25 (-1.35)	
IT Industry	0.04 (0.61)	-0.15 (-1.62)		0.14 (1.31)	-0.05 (-2.48) ^b		0.09 (0.66)	-0.19 (-1.31)	
Medical/Bio-Tech Industry	0.23 (2.37) ^b	0.00 (0.01)		0.30 (1.85) ^c	0.52 (2.67) ^a		-0.03 (-0.22)	0.50 (2.00) ^b	
1998 to 3/2000 Dummy	0.06 (0.74)	0.23 (2.03) ^b		0.07 (0.62)	0.00 (-0.23)		-0.16 (-1.33)	0.26 (1.65) ^c	
Post 3/2000 Dummy	-0.02 (-0.29)	0.12 (1.16)		0.06 (0.59)	0.00 (0.90)		-0.03 (-0.46)	-0.05 (-0.54)	
California Dummy	-0.06 (-1.02)	-0.09 (-1.53)		-0.05 (-0.43)	0.00 (-1.07)		0.04 (0.55)	-0.03 (-0.27)	
Brobeck Investor Dummy	-0.01 (-0.17)	0.11 (1.38)		-0.11 (-0.93)	0.59 (3.75) ^a		77	91	
Observations	147	182		70	91				

* a, b, and c denote significant differences from zero at the 1%, 5%, and 10% levels, respectively.

Table V: Summary Statistics of Outcomes for Firms in Sample

The table displays mean characteristics for various dummy variables associated with firm outcomes. *IPO* and *Acquisition* are one if a liquidity event occurred either as an IPO or as an acquisition, respectively (from archival sources, primarily press releases). *Non-exited survival* is one if the firm was still in business as an independent private entity in March 2008 (from World Wide Web presence and archival sources). *Active non-exited survival* is one if the firm additionally issued at least one press release or updated its website between January 2005 and March 2008. *Inactive non-exited survival* is one if there was neither a press release nor an update to the website during this period. Firms are tagged as *Failure* if they did not experience a liquidity event and they are not still an ongoing private independent firm. *Ex post VC Financing* is one if the firm secures later financing (after Series A) from venture capital firms as reported by Venture Economics. The Venture Economics sample includes US-based venture capital investment targets founded after 1967 with the first investment between 1993 and 2002. Private Equity disbursements labeled as buyout, acquisition, other or unknown were eliminated. Outcome variables of the Venture Economics sample are as reported in that database, but weighted to reflect the distribution of deal origination dates in the Brobeck sample.

Variable	All Deals	Small Deals	Large Deals	Angel-Only Deals	VC-Only Deals	Mixed Deals	Venture	
							Economics	Deals
IPO	0.044	0.022	0.066	0.031	0.105	0.027	0.057	
Acquisition	0.264	0.231	0.297	0.188	0.316	0.268	0.171 ^a	
Non-exited survival	0.280	0.264	0.297	0.500 ^a	0.211	0.241		
Active non-exited survival	0.187	0.154	0.220	0.250	0.184	0.170		
Inactive non-exited survival	0.093	0.110	0.077	0.250 ^a	0.026	0.071		
Failure	0.412	0.484 ^a	0.341 ^a	0.281 ^b	0.368	0.464		
Ex post VC financing	0.489	0.429	0.549	0.281 ^a	0.474	0.554	0.726 ^a	
# Firms in Group	182	91	91	32	38	112	9,902	

* a, b, and c denote significant differences in means at the 1%, 5%, and 10% levels, respectively. The footnotes depicted in columns 2 and 3 are based on tests of differences in means across small versus large firms (column 2 vs 3). The footnotes depicted in column 4 (angel-only) are based on tests of angel-only deals versus mixed deals (column 4 vs column 6). The footnotes depicted in column 5 (VC-only) are based on tests of VC-only deals versus mixed deals (column 5 vs column 6). The footnotes in column 7 (Venture Economics Deals) are based on tests of All Deals (from Brobeck) versus the Venture Economics Deals (column 1 vs column 7).

Table VI: Outcomes versus Deal Characteristics

The table displays Probit model marginal effects, with robust t -statistics in parenthesis. All independent variables (except dummy variables) are standardized to clarify exposition. The dependent variable in the first probit model (columns 1, 3, 6) is a dummy variable indicating whether the firm experienced an Acquisition or IPO by March 2008. The second model's (columns 2, 4, 7) dependent variable is a survival variable which equals one if the firm went public, was acquired, or is still an ongoing independent private firm as of March 2008. The activity adjusted model in column 5 is similar to the survival model in column 4, except that surviving firms that have neither made any press releases nor updated their website between January 2005 and March 2008 have been re-categorized as failures. The independent variables include various investor and firm characteristics, as well as industry and time dummies, and are described in Tables I and II.

Independent Variables	All Deals			Small Deals			Large Deals		
	Probit Model	Probit Model	Probit Model	Probit Model	Probit Model	Probit Model	Probit Model	Probit Model	
	Dep. Var. = Merger/IPO	Dep. Var. = Survival	Dep. Var. = Merger/IPO	Dep. Var. = Survival	Dep. Var. = Survival	Dep. Var. = Survival	Dep. Var. = Merger/IPO	Dep. Var. = Survival	
Angel-Only	-0.04 (-0.37)	0.33 (3.11) ^a	-0.04 (-0.37)	0.43 (2.81) ^a	0.17 (1.22)		0.46 (2.64) ^a	0.18 (1.33)	
VC-Only	0.15 (1.39)	0.18 (1.63)	-0.04 (-0.32)	0.22 (1.12)	0.09 (0.47)		0.18 (1.59)	0.18 (1.61)	
Log Deal Size	0.12 (2.59) ^a	0.18 (3.44) ^a	0.01 (0.18)	0.24 (2.24) ^b	0.20 (2.17) ^b		-0.04 (-0.66)	-0.05 (-0.85)	
Fraction Sold	-0.09 (-1.97) ^b	-0.04 (-0.82)	-0.15 (-2.04) ^b	-0.02 (-0.23)	-0.11 (-1.23)		-0.02 (-0.17)	-0.04 (-0.52)	
Log # Investors	0.00 (0.00)	0.08 (1.40)	0.00 (0.09)	0.15 (1.71) ^c	0.12 (1.65) ^c		-0.30 (-1.95) ^c	-0.48 (-2.01) ^b	
Strategic Alliance	-0.11 (-1.01)	-0.42 (-2.78) ^a	-0.03 (-0.17)	-0.53 (-2.23) ^b	-0.25 (-1.32)		0.56 (2.48) ^b	0.35 (1.88) ^c	
Product Release	0.18 (1.33)	0.30 (2.27) ^b	0.04 (0.22)	0.31 (1.53)	0.09 (0.43)		0.04 (0.57)	0.03 (0.37)	
Log One Plus Firm Age	0.01 (0.35)	0.05 (1.02)	-0.02 (-0.49)	0.05 (0.78)	-0.02 (-0.28)		-0.05 (-0.94)	0.00 (-0.03)	
Log Number of Founders	0.01 (0.14)	0.01 (0.13)	0.03 (0.62)	0.01 (0.12)	0.02 (0.28)		0.22 (1.51)	0.03 (0.23)	
Pre Series A Common Round	0.02 (0.25)	0.00 (-0.03)	0.02 (0.11)	-0.11 (-0.67)	-0.02 (-0.13)		0.00 (0.01)	0.02 (0.13)	
Brobeck Investor Dummy	-0.08 (-0.93)	-0.23 (-2.07) ^b	-0.13 (-1.15)	-0.37 (-2.18) ^b	-0.38 (-2.84) ^a		-0.13 (-0.54)	0.17 (0.83)	
IT Industry	-0.08 (-0.79)	-0.14 (-1.27)	-0.18 (-1.52)	-0.28 (-1.79) ^c	-0.19 (-1.22)		-0.40 (-2.04) ^b	0.22 (1.04)	
Medical/Bio-Tech Industry	-0.23 (-2.02) ^b	-0.10 (-0.57)	-0.14 (-1.11)	-0.29 (-1.32)	-0.19 (-1.03)		0.21 (0.85)	0.02 (0.10)	
1998 to 3/2000 Dummy	-0.13 (-1.24)	-0.13 (-0.98)	-0.15 (-1.41)	-0.25 (-1.49)	-0.35 (-2.47) ^b		-0.04 (-0.20)	0.12 (0.67)	
Post 3/2000 Dummy	-0.25 (-2.38) ^b	0.01 (0.06)	-0.28 (-2.49) ^b	-0.09 (-0.44)	-0.26 (-1.67) ^c		0.45 (3.41) ^a	0.13 (1.22)	
California Dummy	0.26 (3.36) ^a	0.25 (2.87) ^a	0.10 (0.83)	0.32 (2.18) ^b	0.05 (0.36)				
Observations	182	182	91	91	91		91	91	

* a, b, and c denote significant differences from zero at the 1%, 5%, and 10% levels, respectively.

Table VII: Propensity to be Angel-Only Financed and Large Deal Outcomes

The table displays the results of two stage probit models examining the relationship between expected angel-only incidence and outcomes. The first stage model (first column) is a probit model fitting the likelihood that deals are financed exclusively by angels. This model is based on the sample of small firms only (where investor and firm matching is presumed to be unconstrained). The second column is a check as to whether the variables in the first stage model also predict outcomes. The remaining four columns are second stage probit models in which the dependent variable is the IPO/Acquisition dummy. In all four columns, the sample includes only large deals. To construct the propensity variable *Angel-Only Propensity* for the sample of large deals in the last four columns (stage 2), we use the fitted stage 1 model in column 1. This is the propensity that the given firm would have chosen angel-only financing if sufficient funding capacity existed. All independent variables (except dummy variables) are standardized to clarify exposition. The independent variables include various investor, firm and deal characteristics, as well as industry and time dummies, and are described in Tables I and II.

Independent Variables	Stage 1 – <i>Small Deals</i>		Stage 2 – <i>Large Deals</i>			
	Probit Model Dep. Var. = Angel-Only (Stage 1)	Probit Model Dep. Var. = IPO/Acquisition (Check)	Probit Model Dep. Var. = IPO/Acquisition (Stage 2)	Probit Model Dep. Var. = IPO/Acquisition (Stage 2)	Probit Model Dep. Var. = IPO/Acquisition (Stage 2)	Probit Model Dep. Var. = IPO/Acquisition (Stage 2)
VC-Only			0.22 (1.08)	0.42 (2.41) ^b	0.21 (0.94)	
Angel-Only Propensity x Mixed Dummy			-1.75 (-3.03) ^a		-1.62 (-1.66) ^c	
Angel-Only Propensity				-0.80 (-1.97) ^b	0.16 (0.20)	
Log Deal Size			0.24 (1.88) ^c	0.23 (2.06) ^b	0.22 (1.60)	
Log # Investors			-0.09 (-0.87)	-0.05 (-0.51)	-0.03 (-0.27)	
1998 to 3/2000 Dummy			0.07 (0.30)	0.11 (0.43)	0.19 (0.69)	
Post 3/2000 Dummy			-0.15 (-0.67)	-0.16 (-0.74)	-0.06 (-0.24)	
Brobeck Investor Dummy			-0.03 (-0.18)	0.00 (-0.02)	-0.01 (-0.08)	
Log Number of Founders			0.02 (0.30)	-0.03 (-0.54)	-0.01 (-0.10)	
Pre Series A Common Round			0.14 (0.88)	0.24 (1.67) ^c	0.16 (0.97)	
Strategic Alliance			-0.33 (-1.37)	-0.35 (-2.64) ^a	-0.31 (-1.16)	
Product Release			0.62 (1.55)	0.63 (3.03) ^a	0.59 (1.41)	
Log One Plus Firm Age			0.00 (0.03)	0.01 (0.19)	0.03 (0.30)	
IT Industry			0.02 (0.10)	-0.05 (-0.20)	-0.03 (-0.17)	
Medical/Bio-Tech Industry			-0.33 (-1.70) ^c	-0.37 (-2.01) ^b	-0.35 (-1.82) ^c	
California Dummy			0.47 (3.18) ^a	0.45 (3.43) ^a	0.48 (3.18) ^a	
Fraction Sold	-0.54 (-1.98) ^b	-0.09 (-1.42)	-0.17 (-2.34) ^b	-0.12 (-1.55)	-0.13 (-1.60)	
Liquidation Flag	-1.11 (-2.88) ^a	0.12 (1.11)				
Redeemable Flag	-1.06 (-1.94) ^c	-0.13 (-1.06)				
% Common BOD Seats	0.55 (0.96)	-0.09 (-0.59)				
Observations	91	91	91	91	91	91

* a, b, and c denote significant differences from zero at the 1%, 5%, and 10% levels, respectively.

Appendix: Does Angel Participation Matter?

An Analysis of Early Venture Financing

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I Data Sources

Our data are derived from the electronic records of the now defunct law firm Brobeck, Phleger & Harrison (Brobeck). Brobeck had a rich history spanning more than seventy-five years of successful practice, and the failure of the giant firm is a signature event in the history of American legal practice. Founded in San Francisco in 1926, Brobeck served corporate clients in California and the western United States. In 1980, the firm opened a satellite office in Palo Alto from which it developed one of the largest law practices representing technology startups in Silicon Valley and elsewhere (it subsequently opened offices in Austin and the East Coast). The Internet boom of the late 1990s led the firm to pursue a “Big Bet, Big Debt” growth strategy that relied upon rapid growth to support increased infrastructure costs. However, when the technology boom quickly reversed, a self-reinforcing cycle of defections and falling revenues pitched the firm into a “death spiral,” with lease obligations and other fixed costs soaring as a percentage of revenue. The firm decided to cease operations in February 2003, and seven months later creditors forced the liquidating firm to seek bankruptcy protection.¹

In partnership with the National Digital Information Infrastructure Preservation Program of the Library of Congress and assisted by a blue-ribbon advisory council and a team of legal and technical experts, one of the authors has focused on preserving a subset of the digital records of the failed firm. On August 9, 2006, Judge Dennis Montali of the United States Bankruptcy Court, Northern District of California, San Francisco Division, recognized the historic value of these materials and authorized the creation of a *Closed Archive* allowing a significant fraction of these records to be saved. The Court Order specified that the Brobeck Closed Archive will be established under the direction of the Library of Congress and directed the Closed Archive to maintain the confidentiality of the digital records while allowing social science research to proceed using an access model substantially similar to that employed by the U.S. Bureau of the Census.

The present work is the result of an experimental project designed to test the feasibility of conducting social science under the proposed user model. According to the court-approved methodology, access is restricted to archivists and scholars who have signed strict non-disclosure agreements. Access takes place in an on-site, non-networked, institutional setting, and only for specific, enumerated purposes that, among other restrictions, protects the anonymity of investors. Only aggregated or redacted data are allowed to leave the secure area. This solution balances the need to safeguard legal confidentiality while still supporting approved scholarly access.

Brobeck represented both new ventures and investors. Under the experimental protocol,

¹Since the firm announced its intention to close, many articles in the legal and business press have looked at the specific reasons for the failure. Kostal (2003) is the most readable account, but the perspective of the bankruptcy trustee is also highly relevant (Murphy et al. 2005).

we analyzed documents related to first round (Series A) investments. To identify a research sample, an initial query was performed on a subset of the Brobeck digital corpus. The subset consisted of approximately 3.7 million digital records which included word processing documents, spreadsheets, and other electronic miscellany. We focus on six categories of Brobeck representations (matters) that had usable electronic documents concerning relevant deals for the purposes of this study.² We identify 182 Series A funding rounds with sufficiently complete electronic records and an indication of deal closure. The earliest deal occurs in 1993 and the latest deal in 2002.

We carefully examine a number of documents for each deal. The Stock Purchase Agreement establishes the basic terms of the sale of stock, including the type and amount of stock being sold, its purchase price, and numerous conditions for closing and other representations. It also identifies the date of closing, and some information regarding the capitalization prior to closing. This document, along with the term sheet, identifies investors involved in the deal in a capitalization table. The Amended and Restated Charter sets forth investor rights including dividend rights, liquidation preferences, redemption rights, voting rights, and conversion rights/anti-dilution protection. The Voting Agreement sets forth the composition, size, and conditions of election of the Board of Directors. Other documents including Co-Sale and First Refusal Agreements, and Investor's Rights Agreements, provide additional information including details regarding additional rights, restrictions, and option plans. The documents in the Brobeck corpus are very similar to the samples provided by the National Venture Capital Association on their website.³

For each firm in our sample, we also collect complete histories and outcomes based on public sources including Lexis-Nexis, Hoovers, SEC filings, the Internet Archive (archive.org), and Thomson Financial's Venture Economics. In particular, we have a record of each firm's internet presence (from 1996 to present) as well as a complete record of every press release and article about the firm in the popular press. This allows us to identify liquidations, bankruptcies, acquisitions, IPOs, and major company milestones such as strategic alliances, product releases, and subsequent VC investments in the firms.

While we provide a detailed description of the companies in our sample in Section III, it is useful to point out at the outset that we can generally categorize these firms as high growth potential ventures. Most deals inhabit the traditional VC domains of information-technology and biotechnology (even those financed exclusively by angels). It is also reasonable to assume that Brobeck selected deals with high-growth potential. Vronsky (2007) finds evidence that

²Each matter was categorized and also contained a short description of the nature of that particular legal representation. The categories for which there were some matters with the words "Series A" in their descriptions were (with the matter counts shown in parenthesis): "Venture Finance/Company Side" (429), "Venture Financing/Investor Side" (264), "Venture Fund Formation" (133), "Other Financing" (109), "General Business and Technology" (79), and "General Corporate Representation" (44).

³See http://www.nvca.org/model_documents/model_docs.html.

leading Silicon Valley law firms, such as Brobeck, certified deals in the same way that VCs certify deals at the time of IPO in the sense of Megginson and Weiss (1991) and Brav and Gompers (1997). Hence Brobeck had both reputational and monetary incentives to ensure that the deals it represented were of high quality.⁴ We also find that VCs investing in deals in our sample were of higher prominence than the general population of VCs, and the deals in our sample performed well relative to the universe of VC-funded ventures, independent of whether they were financed by VCs, angels, or both.

II Investor Composition and Deal Size

We classify participants in the Series A deals based on three major categories – founders, VCs, and angels. Founders are identified using a two stage process. First, founders are often explicitly identified in the records. When founders are not explicitly identified, but common shareholders are, we identify founders using the following three step rule: (1) the largest common shareholder is identified as a founder; (2) any other common shareholder holding at least 30% as many shares as the largest common shareholder is also identified as a founder; and (3) any common shareholder holding the position of president or CEO is also identified as a founder. Founders ubiquitously hold common shares. For the 165 companies in our sample for which we have founder data, there were a total of 458 founders, leading to an average of 2.78 founders per company, and a range of 1 founder (for 56 companies) up to a maximum of 22 founders in one company (only four companies have more than 8 founders).

We identify venture capitalists primarily by cross-referencing investor names with investors appearing in the Venture Economics database, but we also label as VCs investors with names sharing a common word root with the term “venture”.⁵ This category includes VC arms of banks and corporations which are included in Venture Economics. In addition, there are a small number of professional angel investment groups in our sample. Although angels are the main source of capital for these groups, we put them into the VC category given that their investment process closely resembles that of VCs. In total, there were 482 different VCs that participated in 150 of the 182 deals in our sample.

There is considerable variation in the literature and in practice regarding the exact definition of angel investors. We label as angels all investors who are not otherwise classified as founders or VCs according to the definitions above. This category, which consists of

⁴Law firms commonly rely on either implicit or explicit contingent payment schemes from entrepreneurial companies they represent (Dzienkowski and Peroni 2002; Vronsky 2007). Leading law firms seek to refer strong deals if they wish to work with prominent VCs (Bernstein 1995).

⁵There were a few occasions where investors had the word “venture” in their names but were not in Venture Economics and were not venture capital organizations in the institutional sense we refer to here. In each such case, their investment sizes were under \$50,000. These small investment organizations are often set up for estate planning purposes. We classified these investors as angels.

2,528 different investors across 144 of our 182 sample firms, is predominantly composed of individuals, who invest either directly in their name, or through trusts and other investment vehicles, such as companies set up by individuals for investment purposes.⁶ Although we cannot accurately identify friends and family, we note that investors with the same name or ethnic origin as founders appear to represent a relatively small proportion of the angels and the overall angel investment in our sample.⁷ We also include as angels universities, governments, and other non-VC entities, including Brobeck itself. Since some of the investors we categorize as angels may be viewed as belonging to distinct classes that should be separated from the stereotypical individual angel investors, we later conduct robustness tests to ensure that these non-individual investors do not have a significant impact on any of our key results, and we find that our results are indeed robust. We provide more detailed information on both the angels and VCs in our sample in Section IV. However, the high-potential nature of deals in our sample, as well as the large average amounts invested by angels, suggest that our sample includes sophisticated angels.

Figure 1 shows the distribution of investor composition for deals in our sample. The histogram in the top panel shows the share of investors who are angels. Thirty-two of our deals (18% of the 182 deals) rely solely on angel investments, 38 (21%) have only venture capital investment, and the other 112 deals (61%) draw on both angel and VC investment, with a reasonably uniform distribution of the mix between the two groups of investors.

Note, however, that the bottom panel of Figure 1, which is based on dollars invested rather than number of investors, indicates that venture capitalists systematically invest more money. This suggests that deals involving VC investments are larger, which is indeed the case. Figure 2 shows the distribution of deal size, i.e., dollars invested in Series A deals, for three investor composition categories: angel-only, mixed, and VC-only deals. Angel-only deals are predominantly smaller deals, with a median of \$1.12 million, relative to the overall median of \$3.5 million for our whole sample of Series A deals. Mixed deals tend to be somewhat larger (median of \$4.49 million) than VC-only deals (median of \$3.53 million). While there are numerous VC-only and mixed deals with investment size below the overall sample median, only 3 of the 32 angel-only deals have investment size larger than the sample median. In further support of Finding 2, the presence of VCs appears to be virtually a necessity for closing large deals. Moreover, these general characteristics of our data suggest that there are five subsamples of our overall data that deserve closer scrutiny,

⁶Wong (2002) formally defines angels as those that are “accredited investors” according to SEC Regulation D, Rule 501. Rule 501 states that accredited investors must have a net worth of over \$1M or annual income of over \$200,000. While most angels in our sample are accredited investors, we do find occasional unaccredited investors participating in deals.

⁷Some researchers such as Fenn and Liang (1998) specifically exclude family and friends from their definition of angels. Our general definition of angels, however, is intended to capture investors who are not professional managers of venture capital.

as they may indicate distinctly different deal types in each category: three subsamples based on investor composition (angel-only, VC-only, and mixed deals), and two subsamples sorted on size (large and small deals).

III Firm Characteristics

Table I reports means (standard deviations in parentheses) for numerous characteristics of the 182 firms in our overall sample, as well as across the five subsamples. We assess the representativeness of our overall sample by comparing the characteristics of our firms against those of 9,901 US-based firms with a first recorded investment occurring between 1993 and 2002 (our sample period) for firms founded between 1967 and 2002 (reflecting founding dates in our sample) that are recorded in the Thomson Financial Venture Economics database. We exclude deals labeled as buyout, acquisition, other or unknown. Out of the 182 deals in our sample, 111 also appear in the Venture Economics database (but without the full information to which we have access). We weight statistics from the Venture Economics sample to reflect the (below described) over-representation of recent deals in our sample.

From Panel A, one can see that our sample has a locational bias consistent with Brobeck’s geographical footprint, with a much higher concentration of deals associated with firms based in California (53% vs 36%) and Texas (21% vs 6%). In the “other states” category, 7% of the Brobeck firms are from Colorado, 8% are from the Northeast Corridor (Pennsylvania, New York and New England), and the remaining 11% are scattered in the Midwest, South, Mid-Atlantic, and Washington State, while over 20% of the Venture Economics deals are from the Northeast Corridor and only 2% are from Colorado. However, there are no statistically significant differences in location across the two size subsamples and the three investor composition subsamples.

We use the Venture Economic Industrial Classifications (VEIC) to identify industries, and classify the 71 deals from our sample that were not also in Venture Economics using information from the Brobeck corpus, the World Wide Web and press accounts. We classify 73% of our sample firms as Information Technology firms, as compared to 69% for the Venture Economics firms. Another 12% of our sample firms are Medical/Health/Life Science companies versus 13% of the Venture Economics firms. The remainder of our firms are either non-high technology or unclassified. Overall, our sample firms are quite representative in terms of the concentration in technology and life science industries found for new ventures tracked by Venture Economics. Furthermore, there are no significant differences in industry classification across the size and investor composition subsamples.

In Panel B of Table I, we show that our sample over-represents deals from the most recent subperiod of our study. Half of our Series A rounds occurred after March 2000 (when

the Nasdaq index began its precipitous drop), as compared to 35% for the subset of Venture Economics firms we examine (with a comparable proportion during the boom period of 1998 to 2000). This may reflect a deliberate surge in Brobeck’s business from 2000 to 2002, or alternatively, an increasing reliance on electronic record keeping. Note also that smaller deals are more concentrated in the earlier periods, which likely reflects the propensity to favor larger deals over time (in part due to the fact that deal size is in nominal dollars). However, we do not find any significant increasing or decreasing trends in the occurrence of angel-only or VC-only deals over time.

With respect to deal size, the mean investment size for Series A deals in our sample is \$6.14 Million, which is statistically smaller than the mean of \$7.15 Million for the Venture Economics firms. While not shown in the table, key percentiles of the deal size distribution for the Brobeck firms (and Venture Economics firms in parenthesis) are: 25th percentile is \$1.49 Million (\$1.5 Million); median is \$3.50 million (\$3.89 Million); and 75th percentile is \$6.96 Million (\$8.0 Million), so there is a marked similarity between the low end of the distribution of deal size within our sample relative to the Venture Economics sample, but the right tail of the Venture Economics sample is longer. As was also shown in Figure 2, angel-only deals are much smaller than mixed deals. There is no statistically significant size difference between VC-only and mixed deals. Because we also have information regarding the fraction of each firm that is sold at the time of the Series A deal, we also can report that the average post-money valuation of firms in our sample is \$14.9 million, as compared to \$24.4 million in Venture Economics. This number is also smaller for angel-only deals (\$6.1 million) versus either VC-only or mixed deals (roughly \$16.5 million for both groups).

In terms of age, our sample is overwhelmingly composed of firms that are true start-ups with very recent incorporation dates. The average age is 1.8 years (we can only identify incorporation dates for 85% of the firms in our sample). In contrast, firms experiencing Series A rounds in Venture Economics over the same time period have an average age of 3.1 years (97% of firms have available data). Hence, it appears that Brobeck represented firms that were somewhat less mature than Venture Economics firms.⁸ In the Venture Economics sample, deal size and valuations are correlated with firm age. Hence the differences in age in the samples is consistent with the differences in deal size and post-money valuations.

⁸To calculate age for companies in the Brobeck sample we first looked to documents in the Brobeck collection, which often included incorporation dates. When dates were absent from the Brobeck records, we consulted public sources such as company websites, the Internet Archive (www.archive.org) and press releases. Finally, we relied on Venture Economics if the firm was venture backed and we were unable to date the company based on other sources. To check if this caused any biases, we compared the age in Venture Economics to that from the other sources for 102 of the companies for which we observe age from both sources. Based on Venture Economics, these companies were approximately six months older at time of financing than based on the other sources. However, this difference is not statistically significant. Hence, biases in the data sources do not explain the age differences between the Brobeck sample and the Venture Economics sample.

We find that angel-only deals are just under two months older than VC-only deals on average (2.1 years vs 1.8 years), and three months older than mixed deals (1.7 years). Thus, the fact that Venture Economics does not track angel-only deals does not explain the higher average age for the Venture Economics firms relative to the firms in our sample. We observe age for 28 of the 32 angel-only deals. At time of funding, the youngest quartile of firms is less than 4.6 months old, the median firm is 9 months old, while the top quartile is over 1.5 years old. The oldest angel-only backed firm is 24 years old, reflecting a skewed distribution. For the 95 mixed deals for which we observe age, the youngest quartile, median, oldest quartile, and maximum age are two months, 6.5 months, 1.2 years and 34.5 years, respectively. Finally, for the 32 venture only deals, these four statistics are 5 weeks, 4 months, 1.75 years and 15 years, respectively. A cautionary note is in order. Based on public sources, it is often not possible to ascertain the date of founding with precision. The Venture Economics database commonly assigns January 1st as a founding date when only the founding year is known. To maintain comparability, we followed this convention.

Our sample is thus quite representative relative to firms with Series A deals in the Venture Economics database in terms of industry, but there are also systematic differences with regards to timing, size, age and location. Biases associated with sample selection could affect our results if, for example, there are systematic differences in control rights between the East and West coasts, as suggested by Gupta (2000), or changes in the structure of deals following the peak in new technology venture creation in 2000. We thus control for firm characteristics in the multivariate regressions we present later, both to ensure that the unique characteristics of our sample do not drive any of our results, as well as to see whether the cross-sectional variation in these characteristics within our sample are systematically related to the structure of deals and company success.

Panel B of Table I also shows three additional firm characteristics for which there are no comparable Venture Economics statistics. We find that most firms did not achieve milestones prior to Series A funding, with only 13% having products and 10% being involved in strategic alliances. These findings are consistent with the young age of our firms. Finally, for the firms in our sample, Brobeck represented the company 62% of the time, almost twice as often as they represented one or more of the investors for the deals in our sample (and this is relatively consistent across our subsamples).

IV Investor Characteristics

We now provide more detailed information regarding the investors and investor composition of our deals. Panel A of Table II shows statistics regarding the pre-Series A ownership distribution. We obtain this information from the Series A capitalization tables, but the

archive also contains explicit information describing pre-Series A financing using common shares in about a quarter of the cases (as shown in the last row of Panel A). Not surprisingly, founders retain most of the ownership of the company (approximately 90%) prior to the infusion of significant capital in the Series A round, with statistically insignificant differences in founder ownership across subsamples. As we report earlier, there are on average 2.78 founders per firm, but Panel A shows that smaller firms and firms that have only angel investors in the Series A round have fewer founders on average. Perhaps smaller scale firms require fewer principals to achieve the firm’s objectives, but it could also be that companies with fewer founders are unable to attract the same level of capital as firms with deeper management teams. As expected, the pre-Series A investment largely comes from angels (they obtain 7.6% of the common shares, compared to 2.9% for VCs).⁹

Panel B of Table II provides information related to the Series A round investors. The first two rows summarize key statistics reflected previously in Figures 1 and 2, most prominently that only 3 of the 91 large deals are angel-only deals, while 29 of the 91 small deals do not involve any VCs.

In terms of percentage ownership, founders sell on average almost half of their firm (46.2%) to investors during the Series A round, and somewhat more in larger deals (56.2%). Founders rarely invest money in their own firms in a Series A round (less than .1% of the total investment). In mixed deals, 23.4% of the investment comes from angels, and the balance from VCs. Finally, Panel B shows that, on average, 12.8 angels invest in angel-only deals, 4.8 VCs invest in VC-only deals, and a total of 14.5 investors participate in mixed-deals. Since angels typically do not have as much capital as VC firms, more angel investors are needed to fund an angel-only round.¹⁰ Mixed deals are larger, and thus should be expected to have more investors.

We find that only 9.3% of angels investing in Series A deals had previously invested in the same companies, though this proportion is twice as high for angel-only deals. It is even more uncommon to see the same VCs participating in both pre-Series A and Series-A rounds for a given company (7.1% on average). Panel B also shows that Brobeck invested in approximately one quarter of the deals it handled (either in common or preferred shares), and more so in the larger deals.

Additional investor characteristics are shown in Panel C of Table II. The post-money

⁹All of the ownership percentages reported are based on shares issued in the seed and Series A round, rather than fully diluted shares that take into account warrants and options outstanding. To ensure that dilution does not have an impact on our regression results reported in the paper, we account for the existence of options and warrants, and for their dilutive effect (e.g., on the fraction of ownership sold in a Series A round), and find that our results are robust to these controls.

¹⁰In our sample, the mean investment by an angel is \$150,375, while the median investment size is \$25,000, thus representing a highly skewed distribution which reflects the diverse set of investors captured in our angel category.

ownership figures follow from the pre-Series A and Series-A ownership fractions of the different investors, together with the fraction of the company sold in the Series A round. On average, founders retain majority ownership of their company following the Series-A round only in angel-only deals and small deals. Of course, as we shall see later, this does not necessarily translate into retaining full control of the firms given that founders hold common shares while outside investors predominantly hold preferred shares with more powerful control rights, including board seats.

Finally, in Panel B, we report proximity measures based on the zip code of investors relative to the zip codes of the corporate headquarters, using an automated Mapquest query (we observe zip codes for investors only in 136 of the deals). We find that investors were generally in similar geographic locales as the firms they invested in: 60% were within 3 hours of driving time from the firms they invested in, and 18% were within the same zip code. Investors are closer to the firm in smaller deals, and most likely to be in the same zip code for angel-only deals.¹¹

Given the restrictions we face in the use of our data, there is limited additional information that we can provide to better characterize the angels in our sample. For instance, we are unable to conduct searches that would provide additional information regarding angels' backgrounds (education, and technical and managerial experience), their prior investment experience, their age and wealth, and their relationships to the founders and to the other angels and VCs that invest with them in the deals. The only finer information we can provide is a summary of angel types, which we show in Figure 3. Approximately 70% of investors appear as individuals in the records documenting the Series A rounds. We believe that most of the 10% of investors that are categorized as "small company" are also individuals, investing through corporate vehicles for tax and estate-planning purposes. Together, these two categories make up over 80% of the dollars invested by angels in the mixed and angel-only Series A deals, and even a larger fraction in the angel-only deals. "Companies" and "others" (which include universities, non-profits, and governments) play relatively minor roles relative to individual investors. Note also that while Brobeck invested in a quarter of the deals, their overall stake (dollar weighted participation) in these companies is very small.

Figure 3 shows that angels who invest on their own are more likely to be individuals (82%) than those who invest alongside VCs (66%), though the difference is much smaller (88% versus 82%) if smaller companies and individuals are bundled together as a single category. We have also separately calculated the mean angel investment in angel-only and mixed deals, which are \$78,016 and \$55,771, respectively, for small deals, and the medians in both cases are very close to the overall angel median investment size of \$25,000 (since there

¹¹We also consider (unreported) probit models examining the likelihood that a given deal will be angel-only and VC-only financed given its other characteristics, and we confirm the univariate results reported in the paper.

are few large angel-only deals, it is not relevant to provide the analogous comparison for large deals). Though we have limited information about the angels in angel-only versus mixed deals, it does not appear that there is any significant difference between angels investing in either of these types of deals.

Since most of the VCs investing in Brobeck deals are included in the Venture Economics database, we can provide a more in-depth comparison of the VCs that invest in the 150 Brobeck deals involving VCs relative to the other VCs tracked by Venture Economics. Table III documents various attributes of these VC firms, and highlights any differences between VCs investing in small versus large deals, or in VC-only versus mixed deals. All data, except the centrality measures, are obtained directly from the Venture Economics database.¹² For each private equity fund that invested in one of our sample deals, we aggregate the fund-level information to the VC firm level (and identify this as a “Brobeck VC Firm”). For the universe of Venture Economics VC firms, we exclude firms whose last investment occurred prior to 1993 or whose first investment occurred after 2002 (i.e., outside of our sample period), which yields 5,585 “Non-Brobeck” VC firms.

Panel A of Table III shows that VCs participating in Brobeck deals are more likely to manage US-based funds, in particular California-based funds. They are also more likely to manage early stage funds, but not seed stage funds. In addition, these private equity managers are more likely to manage VC funds and less likely to manage buyout funds. Private equity firms investing in VC-only deals are more likely to manage US funds and with slightly less of an early stage VC focus, as compared to those investing in mixed deals in our sample (while VCs in small and large deals are fairly similar across all these dimensions). We find that Brobeck VC firms are older, and particularly those investing in VC-only deals.¹³

Because age is likely related to long-term success as well as greater VC experience, it appears that VC firms investing in Brobeck deals have higher prestige. Panel B of Table III includes other variables that may proxy for prestige. For example, we find that VC firms that invest in Brobeck deals have a higher incidence of IPOs and acquisitions across all of the companies they invested in through all of their funds.¹⁴ We also find that Brobeck PE firms have raised more venture funds, have managed more capital (per fund and overall), and have a higher centrality measure. While VCs participating in large vs small Brobeck

¹²We thank Mikolaj Jan Piskorski for sharing centrality measures based on Venture Economics data. Piskorski and Anand (2007) calculate eigenvector centrality (Bonacich 1972) through 1999. We matched Piskorski and Anand’s 1999 centrality measures to VC firms in our sample.

¹³We have also conducted a multivariate analysis predicting whether a VC firm becomes affiliated with a Brobeck Series A deal, and this analysis yields similar conclusions to those indicated by the univariate statistics.

¹⁴Ivanov, Krishnan, Masulis, and Singh (2008) find a positive relationship between IPO rates and VC reputation.

deals do not differ across these dimensions, those investing in VC-only deals in our sample have higher success through liquidity events, more funds under management, and more total capital raised than those in mixed deals. Overall, this suggests that VC firms investing in Brobeck deals, and particularly in deals where there are no angels, appear to have high prestige. This is consistent with our belief that Brobeck’s reputation allowed it to attract higher quality clients – both more prominent VCs, and higher quality start ups.

V Deal Characteristics

We turn now to specifics about deal structure, including the types of securities issued and the terms involved. Preferred stock was sold in all but three of the Series A rounds in our sample. Table IV shows that warrants were also sold in 15% of the rounds, and more so in the smaller deals (20%). Employee option plans were set up in 69% of the rounds, and were more prevalent in the large deals (80% of the time) than in the small deals (only 57% of the time), perhaps due to the need to attract a larger and more diverse talent pool for these larger deals. Interestingly, we observe multiple within-round closings in 45% of the deals. This phenomenon, in which investors purchase more shares of the company at identical terms over a period of time, is referred to by Kaplan and Stromberg (2003) as “ex-ante staging”, and appears to be much less prevalent for VC-only deals. This might be due to the tighter control held by the few VC investors in these deals, and the deeper pockets of these VCs, allowing them to commit to immediate funding along with stronger terms. This might also lead to a more easily structured subsequent Series B round as soon as it is required and merited. We find that the average time between first and second closings is 153 days, though it is much longer for large deals (198 days on average).¹⁵

One of the interesting facets of our data is that we are able to observe the number of hours billed by attorneys for each deal. Panel A of Table IV shows that the Series A deals resulted in an average of 169 billed hours per deal. We find that larger deals lead to more billed hours, as do mixed deals (and VC-only deals) relative to angel-only deals. Since one might not expect that the mere scale of a deal should be related to the amount of legal work, we surmise that the higher billed hours of larger deals may reflect that these deals are more likely to involve VCs, who may demand more complex contract terms and be more persistent in negotiating the terms of the deal.

To explore this issue further, we perform (unreported) multivariate regressions of (the log of) billed hours against a number of factors that might contribute to the complexity of the deal, such as investor composition, number of investors, fraction of the firm sold,

¹⁵ Although this phenomenon is interesting, we can report that the presence of multiple closings correlates little with outcomes.

presence of warrants and option plans, number of closings, and time, industry and location dummies. We do find evidence that including warrants and having multiple closings leads to a higher number of billed hours, as does the presence of VCs in the deal, consistent with the notion that more complex and/or contentious deals lead to higher billed hours. However, we also find that size continues to be significant even after controlling for these factors, at least within the subsample of larger firms. This suggests that there may be some other determinants of complexity that are also related to size, perhaps including greater potential reputation and legal risk for a law firm when dealing with larger companies (otherwise, there are less benign explanations for why larger clients are associated with higher billed hours).

We now turn to documenting the relationship between investor composition and cashflow and control rights. In Panel B of Table IV, we summarize the rights associated with the Series A preferred stock based on a review of the closing documents for all the firms in our sample. Gompers (1997) and Kaplan and Stromberg (2003) document that preferred stock is typically differentiated from common stock through superior cashflow rights, voting rights, board representation, liquidation rights, redemption rights, and anti-dilution provisions. Moreover, investment deals are often supplemented by a requirement that the founder's stock be subject to vesting requirements. Consistent with these existing studies, we find substantial variation in the existence and extent of cash flow and control rights, particularly liquidation preferences, redemption rights, cumulative dividend rights, and seats on the board of directors.¹⁶

We classify board seats as being assigned to common shareholders using a two step procedure. First, in many cases, the documents identified which board seats were to be designated by common shareholders or Series A shareholders. Second, for cases in which seat ownership was not specified by share class, but individuals were, we used a fuzzy name matching algorithm to link specific board members to specific investors. As shown in Table IV, common and preferred shareholders have roughly the same representation (46% versus 54%) on the boards of companies across our whole sample. Not surprisingly, firms with VC-only and mixed Series A deals have more board seats assigned to preferred shareholders, as do larger deals, likely due to the larger size and higher concentration of VC participation in those deals.

Turning to cashflow rights, preferred shareholders sometimes have stronger residual cashflow claims in the form of cumulative dividend rights as opposed to regular dividend rights. With regular dividends, an annual payment, often a percentage of investment (generally 8%), is paid conditional on a positive shareholder vote. With cumulative dividend rights, this amount accumulates each year. The cumulative dividend clause is included in relatively

¹⁶Practitioners classify these terms as investor friendly, entrepreneur friendly or neutral (Wilmerding 2003).

few Series A deals (9% of our deals), and never appears when the deal has only angel investors. Cumulative dividends are seen as an investor-friendly term, and provide a strong incentive for the firm to accelerate to a successful exit event. We find that the term is most prevalent in VC-only deals, consistent with the notion that VCs are less patient investors and that they can exert stronger influence on deal terms when they invest on their own.

The variable *Liquidation* is a dummy variable indicating whether preferred shareholders have special liquidation cashflow rights going beyond their initial investment. A value of zero indicates that, after preferred shareholders receive their initial investment, all remaining proceeds upon liquidation go to common shareholders. The dummy variable *Cap on Common*, which takes a value of one for only two deals, indicates that common liquidation amounts are capped. When the *Liquidation* dummy takes a value of one, preferred shareholders have cashflow rights beyond their initial investment, and in all cases but two, they share these additional cashflows equally with common shareholders (in the two cases, all remaining proceeds go to preferred shareholders up to a specified cap). The mean liquidation dummy of 0.42 indicates that 42% of our sample deals provided strong liquidation rights to Series A investors. The dummy variable *Cap on Preferred*'s mean of 0.47 indicates that 47% of these stricter deals had an upper limit on the amount that can be paid to preferred shareholders. Because many preferred liquidation rights are capped, it is important to note that when the company value upon liquidation is sufficiently high, preferred stockholders waive their liquidation rights, and convert their stock to common. Note that the liquidation preference specified in angel-only deals is much less favorable towards the investors, indicating that while angels enjoy some benefits of ownership of preferred, rather than common, shares, their cashflow rights are still relatively weak.

Finally, we find that preferred shareholders have the right to redeem their shares at will in about one quarter of our deals, typically after a period of time and usually conditional on a Series A majority or super-majority vote. Such a right would be invoked when a firm is not performing well, and is considered to be an investor-friendly term. Angels investing on their own seldom demand such a redemption right in their stock purchase agreements.

Given that Kaplan and Stromberg (2003) (KS) examine cashflow and control rights for a similar number of deals, but from a different source and an earlier time period, it is useful to provide a quick comparison of the terms of our respective deals. Our samples differ in important respects. Our data represent 182 series A investments in 182 portfolio firms made by 482 distinct venture capital firms. In contrast, KS analyze 213 investments (of which 98 are series A) in 119 portfolio firms made by 14 VC firms and their affiliates (KS do not report the total number of distinct VCs who invested in the deals in their sample). While they do not distinguish between pure-VC and mixed deals, KS find that non-VCs own, on average, a 20% (non-diluted) stake in investment targets following series A rounds, which

compares to the 16% for mixed deals in our sample. In general, the deals in our sample involve weaker control right provisions.¹⁷ For example, we find that only 8.8% of deals involve cumulative dividend rights, which is considerably below the frequency of 43.8% in KS. We also find that 23.6% of our deals have redemption features, compared to 78.7% reported in KS. (Interestingly, KS reported that 12.9% could redeem shares at fair market value, which is only slightly lower than the 16.4% in our sample). Finally, we find that 85% of our deals include anti-dilution provisions, with the weighted average method used in 92% of those cases, similar to the 95% of deals in KS that include anti-dilution protection, and the 78% of them that use the weighted average method.¹⁸ The differences between our respective samples likely reflect the stage of the deals we analyze, and the presence of angel-only deals in our sample.¹⁹

¹⁷It is difficult to compare the success rates of the two samples, as KS do not report outcomes as hazards which would allow comparison of the likelihood of a given outcome.

¹⁸We do not focus on anti-dilution provisions in our study given that there is little cross-sectional variation across the deals.

¹⁹Gompers (1997) notes certain characteristics of his sample of fifty VC private placement agreements. These appear consistent with our sample. For instance, 51.4% of board seats are controlled by VCs, which is between the 47.8% we find for the VC-only deals and the 60.2% we find for mixed deals in our sample. However, redemption rights are found in 68% of the deals in Gompers' sample, which is more in line with KS than with our sample, again likely reflecting that their sample includes more later-stage rounds.

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Figure 1: **Angel share of deals by number of investors and dollars.**

The figure depicts how deals vary in their investor composition. Top panel: frequency of different investor compositions (fraction of angel investors in deal), unweighted by investment amounts. Bottom panel: frequency of different investor compositions weighted by dollars invested. Both graphs are based on the entire sample (182 firms).

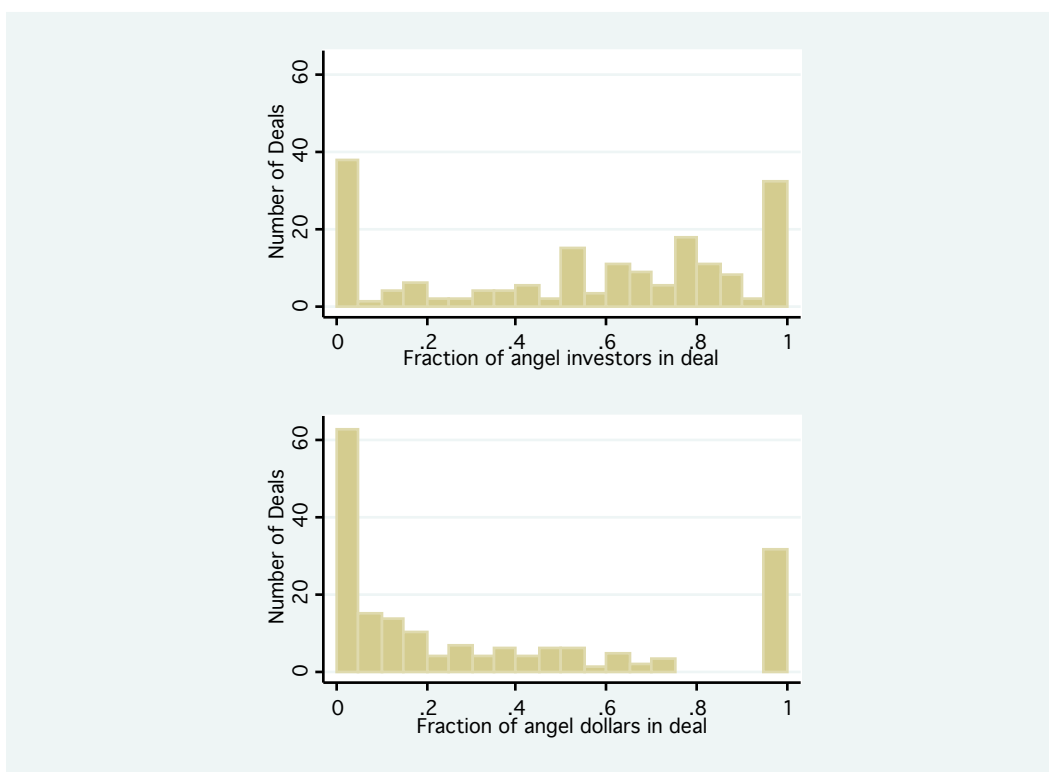


Figure 2: Investment size distribution by investor composition

The figure depicts the distribution of investment size for 32 deals where only angels participated (angel-only deals), 111 deals where both angels and VCs participated (mixed deals) and 38 deals in which only VCs participated (VC-only deals). The size of the deal is measured in millions of dollars. The distribution of the 20th, 40th, 60th, 80th and 90th percentiles are noted by 20, 40, 60, 80 and 90 respectively. The number of deals of magnitude above or below the 50th percentile of all deals for each of the investor composition groups appears to either side of the median deal size vertical line.

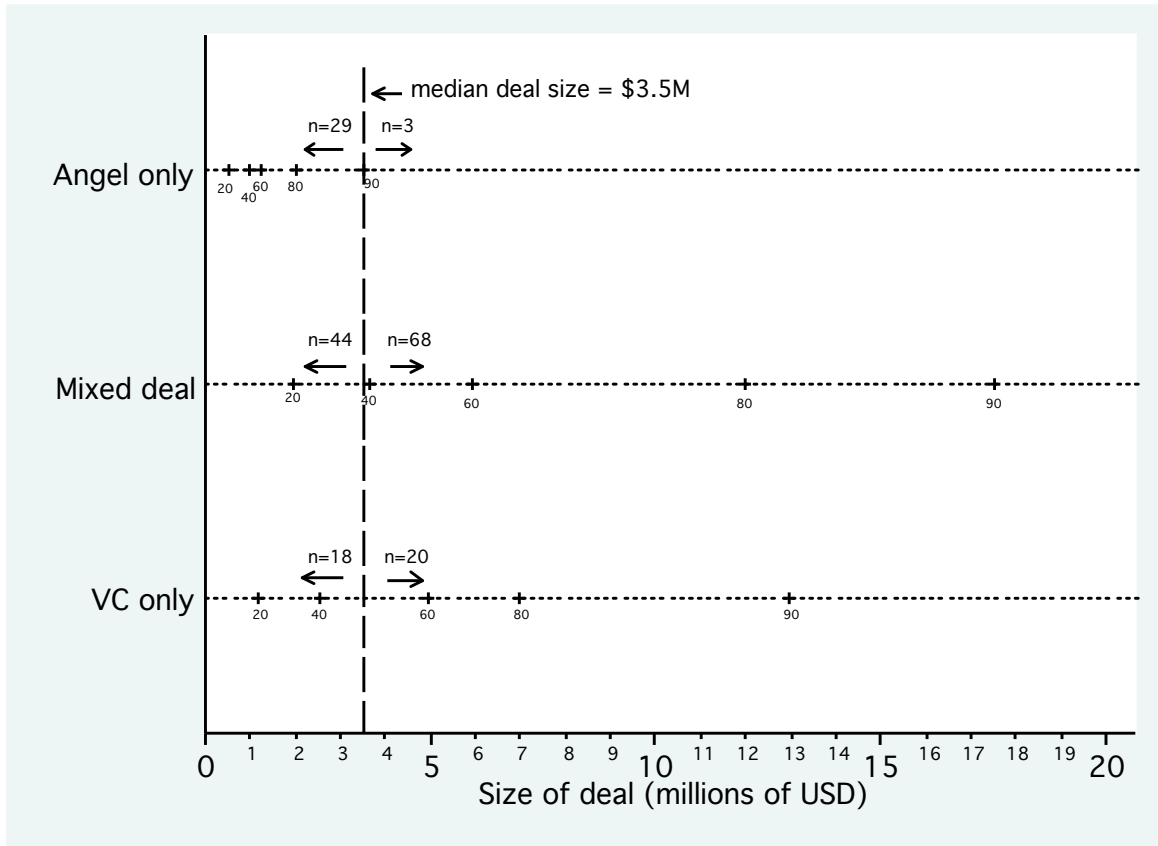


Figure 3: **Composition of Angels**

The figure depicts the average frequency of angel investor types using a deal weighted average (over all deals having at least one angel, angel-only deals, and mixed deals) and a dollar weighted average over all deals having at least one angel. A company is an investor that is incorporated. A small company is an investor that is not incorporated, for example, investors containing the suffixes LLC, LLP or LTD. Angels categorized as others include universities, non-profit organizations, government, and institutions. Unknown are ambiguous investors that we were not able to classify.

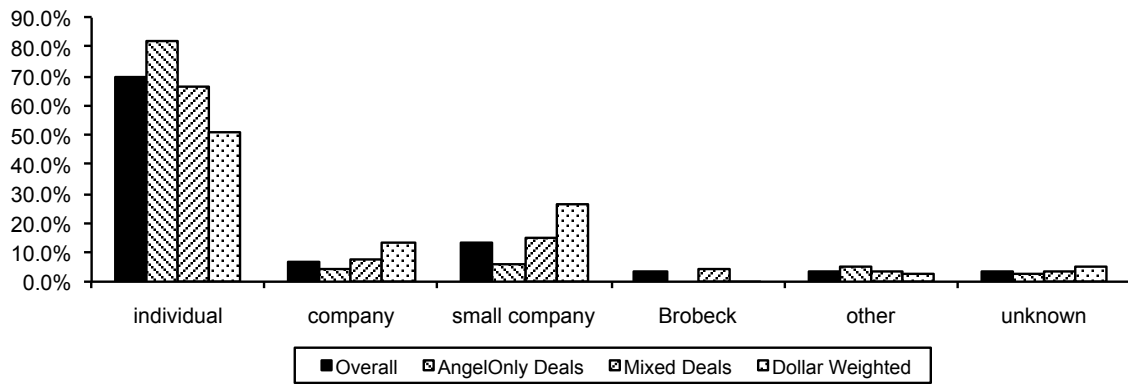


Table I: Summary Statistics for Firms in Sample

The table displays mean characteristics with standard deviations in parentheses. Demographic information shown in Panel A includes location (state dummies which are based on location of company headquarters) and industry classifications, based upon Thomson Financial Venture Economic's proprietary industrial codes (VEIC). *IT Firm* indicates an information technology firm. *Medical/Bio-Tech Firm* indicates firms classified as "Medical/Health/Life Sciences." For deals that did not appear in Venture Economics, classifications were determined according to Brobeck records and public archival sources. Firm characteristics shown in Panel B, such as time period dummies, are based on the date of first closing. *Size* (and its natural log *Log Size*) is measured based on investment dollars, while *Post-Money Valuation* reflects valuation subsequent to the investment in the firm. *Firm Age* is the number of years between the firm's founding date and the date its Series A preferred stock issue closes (computed as number of days divided by 365.25). *Strategic Alliance* and *Product Release* indicate whether the firm issued a press release describing an alliance or product prior to the observed round. *Company matter* takes the value of 1 if Brobeck did not label the representation "Venture Financing/Investor Side." The Venture Economics sample includes US-based private equity investment targets founded after 1967 with the first investment between 1993 and 2002. Private Equity disbursements labeled as buyout, acquisition, other or unknown were eliminated. Statistics of the Venture Economics sample are weighted to reflect the distribution of deal origination dates in the Brobeck sample, period dummies notwithstanding.

Variable	Panel A: Demographics						Venture
	All Deals	Small Deals	Large Deals	Angel-Only Deals	VC-Only Deals	Mixed Deals	Economics Deals
California Dummy	0.529 (0.50)	0.587 (0.49)	0.479 (0.50)	0.565 (0.50)	0.423 (0.50)	0.552 (0.50)	0.360 (0.48) ^a
Texas Dummy	0.206 (0.40)	0.190 (0.39)	0.219 (0.41)	0.217 (0.42)	0.231 (0.43)	0.195 (0.39)	0.060 (0.24) ^a
Other States Dummy	0.265 (0.44)	0.222 (0.41)	0.301 (0.46)	0.217 (0.42)	0.346 (0.48)	0.253 (0.43)	0.581 (0.49) ^a
IT Industry	0.731 (0.44)	0.670 (0.47) ^b	0.791 (0.40) ^b	0.750 (0.44)	0.658 (0.48)	0.750 (0.43)	0.692 (0.46) ^a
Medical/Bio-Tech Industry	0.115 (0.32)	0.121 (0.32)	0.110 (0.31)	0.156 (0.36)	0.158 (0.37)	0.089 (0.28)	0.127 (0.33)
Panel B: Firm Characteristics at time of Series A financing							
Pre 1998 Dummy	0.165 (0.37)	0.231 (0.42) ^a	0.099 (0.30) ^a	0.156 (0.36)	0.211 (0.41)	0.152 (0.36)	0.272 (0.44) ^a
1998 to 3/2000 Dummy	0.341 (0.47)	0.429 (0.49) ^a	0.253 (0.43) ^a	0.406 (0.49)	0.289 (0.46)	0.339 (0.47)	0.384 (0.49)
Post 3/2000 Dummy	0.495 (0.50)	0.341 (0.47) ^a	0.648 (0.48) ^a	0.438 (0.50)	0.500 (0.50)	0.509 (0.50)	0.345 (0.48) ^a
Size (Series A Millions)	6.139 (7.92)	1.554 (0.90) ^a	10.72 (9.11) ^a	1.633 (1.53) ^a	5.756 (8.93)	7.556 (8.19)	7.15 (12.15) ^{b,*}
Log Deal Size	14.98 (1.20)	14.02 (0.77) ^a	15.94 (0.66) ^a	13.94 (0.87) ^a	14.98 (1.07)	15.27 (1.16)	15.08 (1.22) [*]
Post Money Valuation (Millions)	14.82 (22.9)	4.951 (3.27) ^a	24.69 (29.1) ^a	6.068 (4.64) ^a	16.37 (23.7)	16.80 (25.1)	24.4 (42.1) ^{a,†}
Firm Age (years)	1.795 (4.23)	1.924 (5.03)	1.674 (3.34)	2.069 (4.48)	1.831 (3.60)	1.702 (4.39)	3.14 (4.7) ^{a,‡}
% Missing Age Data	0.148 (0.35)	0.176 (0.38)	0.121 (0.32)	0.125 (0.33)	0.158 (0.37)	0.152 (0.36)	0.032 (0.18) ^a
Strategic Alliance	0.104 (0.30)	0.066 (0.25) ^b	0.143 (0.35) ^b	0.031 (0.17) ^a	0.105 (0.31)	0.125 (0.33)	
Product Release	0.126 (0.33)	0.110 (0.31)	0.143 (0.35)	0.125 (0.33)	0.158 (0.37)	0.116 (0.32)	
Company Matter	0.621 (0.48)	0.648 (0.48)	0.593 (0.49)	0.719 (0.45)	0.500 (0.50)	0.634 (0.48)	
# Firms in Group	182	91	91	32	38	112	9901

* a, b, and c denote significant differences in means at the 1%, 5%, and 10% levels, respectively. The footnotes depicted in columns 2 and 3 are based on tests of differences in means across small versus large firms (column 2 vs 3). The footnotes depicted in column 4 (angel-only) are based on tests of angel-only deals versus mixed deals (column 4 vs column 6). The footnotes depicted in column 5 (VC-only) are based on tests of VC-only deals versus mixed deals (column 5 vs column 6).
^{*} 9,345 observations; [†] 3,791 observations; [‡] 9,498 Observations

Table II: Summary Statistics for Investors

The table displays mean characteristics with standard deviations in parentheses. Pre Series A investor characteristics are in Panel A. *Pre Series A Founder, Angel, and VC Ownership* break down the components of initial ownership in the company. *Pre Series A Common Round* is a dummy variable that is one if a pre Series A financing based on common stock is observed. Series A investor characteristics are in Panel B. *Angel-Only (VC-only)* are dummy variables equal to one if all investors in the deal are angels (VC firms). *Fraction Sold* shows the percentage of the company sold to investors in the Series A round. *% Founder Series A, % Angel Series A* and *% VC Series A* break down investment in the Series A round. *# Investors* is a count of investors in the deal. Repeat investments by the same investor (generally across multiple closings) are counted as a single investor (*Log # Investors* is the natural log.) *Repeat Angel Flag* takes the value of 1 if an angel who invests in the Series A round had previously invested in the firm. The *Brobeck Investor Dummy* is one if Brobeck invested in the firm (either common or preferred shares). *% Investors Within 3 Hours* is the share of investors (unweighted by dollars invested) who are within three hours driving distance. Distance is measured from zip-code center to zip-code center using an automated Mapquest query. *% Investors in Same Zip Code* is the share of investors (unweighted by dollars invested) in the same zip code. In Panel C, *Final Angel Ownership, Final Founder Ownership, and Final VC Ownership* represent the respective ownership shares post-money, i.e., after the Series A round closes (before any subsequent dilution from options).

Variable	All Deals	Small Deals	Large Deals	Angel-Only Deals	VC-Only Deals	Mixed Deals
Panel A: Pre Series A Investor Characteristics						
Pre Series A Founder Ownership	0.895 (0.21)	0.902 (0.16)	0.888 (0.26)	0.927 (0.12)	0.832 (0.35)	0.907 (0.17)
Pre Series A Angel Ownership	0.076 (0.15)	0.076 (0.14)	0.076 (0.16)	0.066 (0.10)	0.091 (0.20)	0.074 (0.14)
Pre Series A VC Ownership	0.029 (0.10)	0.022 (0.08)	0.036 (0.12)	0.007 (0.04)	0.077 (0.19) ^b	0.018 (0.06)
Number of Founders	2.775 (2.46)	2.374 (1.54) ^a	3.176 (3.08) ^a	1.969 (1.06) ^a	2.895 (3.64)	2.964 (2.22)
% with Founder Data	0.907 (0.29)	0.890 (0.31)	0.923 (0.26)	0.844 (0.36)	0.921 (0.27)	0.920 (0.27)
Pre Series A Common Round	0.253 (0.43)	0.286 (0.45)	0.220 (0.41)	0.219 (0.42)	0.263 (0.44)	0.259 (0.44)
Panel B: Series A Investor Characteristics						
Angel-Only	0.176 (0.38)	0.319 (0.46) ^a	0.033 (0.18) ^a	1.000 (0.00)	0.000 (0.00)	0.000 (0.00)
VC-Only	0.209 (0.40)	0.198 (0.40)	0.220 (0.41)	0.000 (0.00)	1.000 (0.00)	0.000 (0.00)
Fraction Sold	0.462 (0.22)	0.363 (0.15) ^a	0.561 (0.23) ^a	0.303 (0.16) ^a	0.436 (0.24) ^b	0.516 (0.21)
% Angel Series A	0.319 (0.36)	0.464 (0.40) ^a	0.174 (0.24) ^a	0.996 (0.01) ^a	0.000 (0.00) ^a	0.234 (0.21)
% Founder Series A	0.001 (0.00)	0.002 (0.01)	0.001 (0.00)	0.004 (0.01)	0.000 (0.00) ^a	0.001 (0.00)
% VC Series A	0.679 (0.36)	0.533 (0.41) ^a	0.825 (0.24) ^a	0.000 (0.00) ^a	1.000 (0.00) ^a	0.764 (0.21)
# Investors	12.12 (13.4)	10.80 (14.4)	13.45 (12.3)	12.81 (20.7)	4.763 (5.62) ^a	14.42 (11.9)
Log # Investors	1.994 (1.06)	1.762 (1.15) ^a	2.227 (0.92) ^a	1.711 (1.30) ^a	1.044 (0.99) ^a	2.398 (0.73)
Repeat Angel Flag	0.093 (0.29)	0.121 (0.32)	0.066 (0.25)	0.188 (0.39)	0.000 (0.00) ^a	0.098 (0.29)
Repeat VC Flag	0.071 (0.25)	0.055 (0.22)	0.088 (0.28)	0.000 (0.00) ^a	0.105 (0.31)	0.080 (0.27)
Brobeck Investor Dummy	0.236 (0.42)	0.105 (0.37) ^a	0.308 (0.46) ^a	0.063 (0.24) ^a	0.079 (0.27) ^a	0.339 (0.47)
% Investors within 3 Hours	0.604 (0.36)	0.657 (0.35)	0.558 (0.37)	0.649 (0.35)	0.514 (0.47)	0.619 (0.32)
% Investors in Same Zip Code	0.179 (0.30)	0.248 (0.36) ^a	0.120 (0.23) ^a	0.302 (0.39) ^a	0.272 (0.40) ^b	0.119 (0.22)
Panel C: Other Characteristics						
Final Angel Ownership	0.173 (0.17)	0.206 (0.18) ^a	0.140 (0.16) ^a	0.377 (0.20) ^a	0.039 (0.09) ^a	0.160 (0.13)
Final Founder Ownership	0.461 (0.23)	0.560 (0.18) ^a	0.362 (0.23) ^a	0.615 (0.20) ^a	0.448 (0.27)	0.422 (0.20)
Final VC Ownership	0.366 (0.27)	0.233 (0.22) ^a	0.498 (0.26) ^a	0.007 (0.04) ^a	0.513 (0.28) ^b	0.418 (0.21)
# Firms in Group	182	91	91	32	38	112

* a, b, and c denote significant differences in means at the 1%, 5%, and 10% levels, respectively. The footnotes depicted in columns 2 and 3 are based on tests of differences in means across small versus large firms (column 2 vs 3). The footnotes depicted in column 4 (angel-only) are based on tests of angel-only deals versus mixed deals (column 4 vs column 6). The footnotes depicted in column 5 (VC-only) are based on tests of VC-only deals versus mixed deals (column 5 vs column 6).

Table III: Summary Statistics for VC Firms in Sample

The table displays mean characteristics with standard deviations in parentheses. The sample includes all private equity firms which made investments between 1/1/1993 and 12/31/2002 that are included in the Thomson Financial Venture Economics database. The statistics include the histories of these private equity firms, including funds that were managed outside the 1993-2002 time frame. Statistics are reported by whether the firm was represented in the Brobeck Sample (columns 1 and 2, conditional on inclusion in Venture Economics), whether the firm was associated with a small or large deal (columns 3 and 4), and whether the firm was associated with a mixed or VC-only deal (columns 5 and 6). All fund statistics are aggregated at the VC firm level. *Share of funds hq in northern CA* and *Share of funds hq in US* represent the share of funds under the private equity firm's management that report headquarters in northern California and the United States respectively. *Share of funds with Seed Stage Focus* and *Share of funds with Early Stage Focus* represent the share of funds under the firm's management that report a focus on seed stage deals or early stage deals. *Share of VC funds* and *Firm's share of buyout funds* represent the share of funds under the VC firm's management reporting that they are venture capital or buyout funds. *# funds under firm management* is the number of funds ever managed under the private equity firm's management. *Log total capital raised* is the natural log of the total capital (in millions of US dollars) raised by these funds. *Log fund size, \$ M* is the mean capital raised for each fund under a private equity firm's management. These latter two variables reflect a smaller sample size than some variables as only 87% of our sample VCs report these statistics. *Age* is measured relative to 2002. *IPO rate*, and *Acquisition rate* are the share of the VC's investment targets that reach the mutually exclusive categories IPO or merged/acquired, respectively. Rates are calculated as the share of investment targets in these outcome categories, as reported in Venture Economics. *Centrality* is eigenvector centrality (Bonacich 1972).

	Brobeck VC firms	Non-Brobeck VC firms	Small deal VCs	Large Deal VCs	VCs in mixed deals	VC in VC-only deals
Panel A: Demographics						
Share of funds hq in northern CA	0.414 (0.49)	0.095 (0.29) ^a	0.403 (0.48)	0.418 (0.49)	0.410 (0.49)	0.428 (0.49)
Share of funds hq in US	0.909 (0.29)	0.639 (0.48) ^a	0.916 (0.28)	0.906 (0.29)	0.891 (0.31)	0.972 (0.17) ^a
Share of funds w/ Seed Stage Focus	0.038 (0.13)	0.061 (0.23) ^a	0.022 (0.10)	0.044 (0.14) ^c	0.043 (0.15)	0.021 (0.05) ^b
Share of funds w/ Early Stage Focus	0.417 (0.40)	0.223 (0.39) ^a	0.417 (0.38)	0.417 (0.40)	0.443 (0.40)	0.326 (0.37) ^a
Share of VC funds	0.935 (0.20)	0.818 (0.37) ^a	0.927 (0.22)	0.938 (0.19)	0.945 (0.20)	0.901 (0.21) ^c
Share of buyout funds	0.035 (0.15)	0.118 (0.31) ^a	0.056 (0.20)	0.029 (0.12)	0.031 (0.14)	0.053 (0.15)
Panel B: Prestige/Prominence						
# funds under firm management	7.658 (6.84)	2.040 (2.18) ^a	7.073 (6.23)	7.858 (7.04)	7.178 (6.74)	9.358 (6.95) ^a
Log total capital raised	5.300 (2.91)	2.482 (2.57) ^a	4.980 (2.82)	5.410 (2.94)	5.137 (2.82)	5.877 (3.17) ^b
Log fund size	3.924 (2.05)	2.191 (2.24) ^a	3.713 (1.94)	3.996 (2.08)	3.841 (2.01)	4.220 (2.17)
Age	13.137 (8.13)	7.041 (6.43) ^a	13.350 (7.97)	13.064 (8.19)	12.431 (8.21)	15.64 (7.31) ^a
IPO rate	0.150 (0.13)	0.118 (0.21) ^a	0.154 (0.16)	0.149 (0.13)	0.141 (0.13)	0.182 (0.13) ^a
Acquisition rate	0.313 (0.17)	0.221 (0.27) ^a	0.322 (0.20)	0.310 (0.16)	0.303 (0.18)	0.347 (0.14) ^a
Centrality	0.074 (0.14)	0.002 (0.02) ^a	0.105 (0.19)	0.063 (0.12) ^b	0.065 (0.14)	0.106 (0.16) ^b
Observations	482	5585	123	359	376	106

* a, b, and c denote significant differences in means at the 1%, 5%, and 10% levels, respectively. These are based on tests of differences in means across Non-Brobeck VC firms and Brobeck VC firms (footnotes displayed in column 2), VCs affiliated with small versus large deals (displayed in column 4), and VCs affiliated with VC-only deals versus mixed deals (displayed in column 6).

Table IV: Summary Statistics for Deals

The table displays mean characteristics with standard deviations in parentheses. *Warrants Dummy* is one if warrants were issued in the Series A Financing. *Options Dummy* is one if an option plan was set up concurrent with the Series A Financing. *# closings* is the number of separate executed closings under the Series A terms (*Multiple Closings* indicates more than one). *Average Days to Second Closing* is the days between the first and second closings. All preceding variables are taken from the electronic record of the closing documents in the Brobeck corpus. *Billed Hours* reflects the total billed hours associated with Brobeck's representation of either the company or its investors in the deal. Control rights in Panel B and terms are extracted from closing documents from the Brobeck corpus. *% Common BOD Seats* is the share of board seats held by common shareholders (from the "voting rights agreement", or when specific parties are named, by cross-referencing with investors). *Cumulative Dividend Flag* is a dummy equal to one if preferred shareholders have a right to dividends that accumulate over the time of their investment. *Liquidation Flag* is a dummy equal to one when preferred shareholders have liquidation rights exceeding the value of their initial investment. *Cap on Preferred* is a dummy equal to one if these additional preferred stock liquidation rights are capped at a multiple of the initial investment. *Cap on Common* is a dummy equal to one if common stock liquidation rights are capped at a multiple of the investment. *Redeemable Flag* is a dummy equal to one when preferred shareholders can demand that the firm repurchase their shares. *Delayed Redemption* is a dummy equal to one if preferred stockholders' redemption rights are time-delayed. *% Series A Req. to Vote* is the percentage of Series A shareholders required to invoke a redemption.

Variable	All Deals	Small Deals	Large Deals	Angel-Only Deals	VC-Only Deals	Mixed Deals
Panel A: Deal Characteristics						
Warrants Dummy	0.154 (0.36)	0.198 (0.40) ^b	0.110 (0.31) ^b	0.250 (0.44)	0.079 (0.27)	0.152 (0.36)
Options Dummy	0.687 (0.46)	0.571 (0.49) ^a	0.802 (0.40) ^a	0.594 (0.49)	0.658 (0.48)	0.723 (0.44)
Multiple Closings Flag	0.445 (0.49)	0.407 (0.49)	0.484 (0.50)	0.375 (0.49)	0.237 (0.43) ^a	0.536 (0.50)
# Closings	1.769 (1.19)	1.736 (1.29)	1.802 (1.09)	1.750 (1.36)	1.553 (1.40)	1.848 (1.05)
Average Days to Second Closing	153.4 (157)	103.7 (89.4) ^a	197.9 (190) ^a	138.2 (93.5)	232.9 (149)	144.5 (166)
Billed Hours	169.0 (115)	147.3 (103) ^a	190.7 (123) ^a	109.1 (70.9) ^a	185.4 (155)	180.6 (105)
Log Billed Hours	4.878 (0.82)	4.737 (0.85) ^a	5.017 (0.76) ^a	4.488 (0.78) ^a	4.901 (0.88)	4.977 (0.78)
Panel B: Control Rights						
% Common BOD Seats	0.464 (0.33)	0.579 (0.33) ^a	0.359 (0.29) ^a	0.675 (0.31) ^a	0.522 (0.39)	0.398 (0.29)
Cumulative Dividend Flag	0.088 (0.28)	0.044 (0.20) ^a	0.132 (0.34) ^a	0.000 (0.00) ^a	0.158 (0.37)	0.089 (0.28)
Liquidation Flag	0.418 (0.49)	0.374 (0.48)	0.462 (0.50)	0.125 (0.33) ^a	0.579 (0.50)	0.446 (0.49)
Cap on Preferred	0.474 (0.50)	0.441 (0.50)	0.500 (0.50)	0.500 (0.57)	0.364 (0.49)	0.520 (0.50)
Cap on Common	0.019 (0.13)	0.035 (0.18)	0.000 (0.00)	0.000 (0.00)	0.000 (0.00)	0.032 (0.17)
Redeemable Flag	0.236 (0.42)	0.209 (0.40)	0.264 (0.44)	0.031 (0.17) ^a	0.368 (0.48)	0.250 (0.43)
Delayed Redemption	0.225 (0.41)	0.209 (0.40)	0.242 (0.43)	0.031 (0.17) ^a	0.368 (0.48)	0.232 (0.42)
% Series A Req. to Vote	57.55 (8.40)	57.12 (8.54)	57.87 (8.52)	50.00 (0.00)	54.16 (7.53) ^a	59.96 (8.35)
# Firms in Group	182	91	91	32	38	112

* a, b, and c denote significant differences in means at the 1%, 5%, and 10% levels, respectively. The footnotes depicted in columns 2 and 3 are based on tests of differences in means across small versus large firms (column 2 vs 3). The footnotes depicted in column 4 (angel-only) are based on tests of angel-only deals versus mixed deals (column 4 vs column 6). The footnotes depicted in column 5 (VC-only) are based on tests of VC-only deals versus mixed deals (column 5 vs column 6).